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The effect of interventions for the wellbeing, satisfaction and flourishing of general practitioners – A systematic review

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TITLE PAGE

Title	The effect of interventions for the wellbeing, satisfaction, and flourishing of general physicians – A systematic review
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ABSTRACT

Objectives

Clinician wellbeing has been recognised as an important pillar of health care. However, research mainly addresses mitigating the negative aspects of stress or burnout, rather than enabling positive aspects. With the added strain of a pandemic, identifying how best to maintain and support the wellbeing, satisfaction and flourishing of frontline doctors is now more important than ever.

Design

We systematically reviewed the effect of any type of intervention on wellbeing, and related positive outcomes in General Practitioners (GPs).

Data sources

We searched MEDLINE, PsycINFO, Embase, CINAHL, and Scopus from 2000 to 2020.

Study selection

Studies with more than 50% GPs, trialling any type of intervention, looking at wellbeing, satisfaction, flourishing and related positive outcomes were included. The Cochrane risk of bias 2 tool was applied.

Results

Data Extraction

We retrieved 14,792 records, 94 studies underwent full text review. We included 19 studies in total. Six randomised controlled trials, three non-randomised, controlled trials, and eight non-controlled studies. There were a total of 1141 participants, with interventions targeting individuals or organisations, plus two quasi-experimental articles evaluating health system policy change.

Data synthesis

Individual mindfulness interventions were the most common (k=9) with medium to large within (0.37-1.05) and between group (0.5-1.5) effect sizes for mindfulness outcomes, and small to medium effect sizes for a range of other positive outcomes including resilience, compassion, and empathy. Studies assessing other interventions or positive outcomes (including wellbeing, satisfaction) were of limited size and quality.

Conclusions

There is remarkably little evidence on how to improve PCP wellbeing beyond using mindfulness interventions, particularly interventions addressing organisational or system factors. This was further undermined by inconsistent reporting, and overall high risk of bias. We need to conduct research in this space with the same rigour with which we approach clinical intervention studies in patients.

Registration

We submitted the systematic review protocol for registration on PROSPERO.

Funding source

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ARTICLE SUMMARY

Strenghts and limitations of this study

- Whilst burnout has been a general focus of research, interventions to improve the wellbeing of general practitioners (primary care physicians) appear sparse.
- The strength of this study is the extensive and systematic approach taken to evaluate interventions aimed at improving wellbeing, satisfaction, flourishing and other positive outcomes in general practitioners
- The systematic review was registered on PROSPERO a priori before commencing the data selection and extraction process.
- The limitations of this systematic review is the dependency on the number of retrieved and included publications, and their quality of methodology and reporting.

INTRODUCTION

Mental ill-health, burnout and stress amongst health care practitioners are a huge concern internationally with impacts on individual doctors and their families, patient care, and the sustainability of the health care system (1). Hence, improving clinician wellbeing has been added to the more commonly shared health system goals of: improved care for individuals, better population health, and reduced health care costs (2-5). Despite this stated aim, few studies have evaluated interventions to improve wellbeing, satisfaction, and flourishing in general practitioners (GPs) (6-12), rather than taking a problem-focused context, such as reduction of burnout and stress.

a) Factors associated with satisfaction and wellbeing in GPs

Primary care physicians are less satisfied with their job than other specialists in the USA (13, 14), a ten-year decline in job satisfaction for British GPs has been reported (15), and a Norwegian longitudinal study reported dwindling GP satisfaction over seven years (16). In primary care, job satisfaction correlates with practice resources, an ordered atmosphere, a practice culture that enables communication, and ease of providing quality care (13, 17, 18), and is inversely related to isolation and low sense of community (19, 20).

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For wellbeing, Murray et al. conducted a cross-sectional study exclusively exploring positive mental health and psychological resources (wellbeing, resilience, self-efficacy and hope) of GPs (21). GPs positive mental health was comparable to the general population and older and female GPs fared best, suggesting interventions for younger and male GPs may be most useful.

b) Systematic reviews of interventions aimed at improving satisfaction and wellbeing in GPs

A European collaboration conducted a systematic review and qualitative study looking at positive determinants of satisfaction in GPs. They identified general work related factors (i.e. workload, responsibility, recognition and income), and GP specific factors (i.e. competence, intellectual stimulus and work-life balance) (22, 23). However, there does not appear to be a systematic review looking at interventions to improve satisfaction exclusively in GPs.

A systematic review of interventions to improve the psychological wellbeing of GPs identified only four articles; two cognitive-behavioural, one mindfulness-based intervention and one self-help information approach (24). They applied a dual model of languishing/flourishing and the presence of mental illness/absence of mental illness matrix (25-28).

Overall, little seems known about which interventions are efficacious in promoting GPs wellbeing and satisfaction. Currently, COVID-19 places enormous additional strain on health professionals which impacts their physical, mental and social wellbeing (29, 30). GPs as the first port of call are particularly affected, whilst playing a crucial role in supporting population health (31-33). Now more than ever, efforts must be made to ensure GPs remain well.

OBJECTIVES

We systematically reviewed studies exploring the effect of any type of intervention on the wellbeing, satisfaction and wellbeing of GPs.

METHODS

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Data sources and search strategy

We followed PRISMA guidelines (34), and consulted a specialist librarian (JKH). MEDLINE, PsycINFO, Embase, CINAHL and Scopus were searched from 01/2000 to 01/2020.

PICO search terms included GPs and synonyms; interventions and implementations directed at the individual and profession at large; outcomes included wellbeing, satisfaction, flourishing, synonyms and antonyms (*search strategy, supplement 1)*. Titles, abstracts, text, key terms and subject headings were searched for English publications. Eligible articles were hand-searched for further relevant references.

Study selection and data extraction

Sample screening of titles was performed by two reviewers (DNA, AS) and showed excellent agreement. DNA screened all titles. DNA and AS independently examined all abstracts and full-text records for inclusion using Covidence (35). Any disagreement was resolved through discussion, or third reviewer adjudication (NG).

Studies with more than 50% GPs (family doctors, primary care physicians) working in a practice setting or medical centre, reporting on wellbeing, satisfaction, flourishing, mindfulness, resilience, empathy, engagement, balance, empowerment, compassion, work-related morale and control measures were included. We excluded studies exclusively reporting on burnout, distress or mental ill-health.

Data including author, year, type of study, participants, intervention, pre- and post-outcome measures, and results were extracted. *(Table 1)*.

Data synthesis and analysis

We calculated within- and between-group absolute change and effect sizes (Hedges' g) *(see table 2, 3 and supplement 2, 3)* (36). We compared mean outcome scores and standard deviations (SDs) at baseline with post-intervention scores. Where several post-intervention measures were reported, the primary outcome point nominated by the authors was selected. We utilised SD*, which takes different sample sizes into account *(formulae in supplement 2)*. For within-group we calculated the pooled SD* based on pre- and post- interventions SDs, for between-group analysis the effect size was

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calculated based on the pooled SD* of control and intervention group at baseline (37) *(supplement 2).*

Positive effect sizes indicated an effect for the intervention. Effect sizes of 0.2, 0.5 and 0.8 were considered small, moderate and large, respectively (38).

Risk of bias

Two reviewers (AS, DNA) independently applied the Cochrane RoB2 (39) to RCTs. Any discrepancies were discussed, and a consensus was achieved. The other studies were assessed by DNA.

Patient and public involvement

No patient are involved.

RESULTS

The database searches rendered 14,792 records in total. After removing duplicates, studies conducted before the year 2000, and adding 12 studies through hand search, 10,759 studies were screened. We eliminated 9,682 records by title, and 983 by abstract, leaving 94 studies for full-text assessment.

Study characteristics & design

We included nineteen studies in the systematic review (2, 10, 40-56) *(Table 1 and 2, and PRISMA-Flowchart Figure 1)*. Six randomised controlled trials (RCT), three non-randomised controlled trials (CBA), eight non-controlled interventions (NCBA), and two reports from a longitudinal cohort during which a health policy change was introduced, which we considered as 'naturalistic' interventions (41, 56). Five studies included a qualitative component. Only one RCT (50) and two CBAs (44, 47) utilised active controls. Five RCTs and one CBA (40) had a waitlist control arm. Publications from the United States (8/19, 42%), Europe (8/19, 42%), Australia (2/19) and Israel (1/19) were included *(Table 1).*

Participants

The total population enrolled was 1,141 for the 17 intervention studies. The two studies reporting on the same panel survey population (41, 56) included approximately 2,000

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participants each year. Mean age ranged from 40 – 54.9 years, and sex from 8% - 76.9% male participants *(Table 1)*. Attrition for intervention groups varied from 0% to 20%, for controls from 0% to 24%. One outlier had a total attrition rate of 80% (51).

Intervention type

We found considerable variation in intervention type, length and dose-intensity. Three groups were distinguished based upon the focus of the intervention: individual/personal (13/19, 68% of studies, N=930), organisational (4/19, 21%, N=211) and naturalistic interventions on a systemic level (policy change) (2/19, 11%).

Individual mindfulness-based interventions were most common (9/13, 69%), followed by educational training or experiential workshops (3/13, 23%) with one coaching intervention (46). Two organisational interventions trialled the addition of clerical support or scribes, and two explored an organisational improvement programme. Two studies from the United Kingdom examined the effects of the introduction of a pay for performance scheme (41, 56) *(Table 2)*.

Outcomes and their measures

The definitions of outcomes and measurement tools varied considerably. Only one study clearly stated one a priori primary outcome (51), with most using a battery of self-reported outcome measures *(supplement 4)*. These included a range of validated and non-validated job satisfaction metrics in eight studies, mindfulness tools in six, resilience measures in four, compassion and empathy tools were each used thrice, the positive and negative affect scale was used twice. The WHO-5 wellbeing index was used once *(supplement 4)*. Not one study evaluated flourishing. Negative outcome measures were often concurrently reported. Sixteen studies employed the Maslach Burnout Inventory or other stress-related measures.

Intervention effects

The between-group change for controlled studies and within-group change for intervention arms are presented in *Tables 2, 3 and supplement 3*.

a. *Individual focussed interventions*

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i) Mindfulness (k=9)

Six mindfulness interventions (3 RCTs, 1 CBA, 2 NCBA) evaluated mindfulness outcomes (FFMQ, MAAS, BAER) and reported moderate to high between-group effect sizes (k=4) ranging from 0.5 – 0.88 for mindfulness with an outlier at 1.5 (42). Within-group ES (k=6) showed moderate effect sizes (range 0.47 – 0.78) with one study outlier at 0.37 and one at 1.05 (*Tables 2 and 3*).

Studies frequently utilised resilience, compassion and empathy measures with overall low to moderate effect sizes. One RCT and two NCBAs measured resilience (BRS, RS-14, CD-RISC), whereby between-group ES (k=1) was moderate at 0.61, while within-group (k=3) effect sizes were low to moderate (range 0.17 – 0.51). Compassion measures (SCBC, SCS) were reported in three studies (1 RCT, 2 NCBAs). Between-group ES (k=1) was 0.73, whilst within-group ES (k=3) varied considerably (-0.04 to 0.77). Three studies reported on empathy (JSPE) (1 RCT, 1 CBA, 1 NCBA) with very low 0.02, respectively moderate between-group 0.44 ES (k=2), and within-group ES ranging from 0.2 - 0.44 (k=3) *(Tables 2 and 3)*.

Two mindfulness studies (NCBAs) measured positive affect (PANAS), only one reported a within-group ES (0.52). One NCBA reported a within-group ES (0.52) for wellbeing (WHO-5), another NCBA reported an ES of 0.46 for self-reflection.

These effect sizes are generally supported by the results reported in the individual studies *(Table 1)*. Several interventions included repeated measures at later time-points , i.e. during maintenance phase (48), showing an ongoing impact of mindfulness practice. Qualitative results suggested increased wellbeing and compassion towards self and others (40), respectively, improved awareness, acceptance, peacefulness and openness (55) after the intervention.

ii. Training, workshops and coaching (k=4)

For training, workshops and coaching interventions, we were only able to obtain data to calculate the ES of one RCT (46) and one CBA (47). Low between-group effect sizes for work-related morale (0.3), quality of work-life (0.27), and low ES for both measures within-group (0.43 and 0.45 respectively) were found for Gardiner's CBA (47). Very low effect sizes for job satisfaction and resilience both for between-group (0.06, 0.13), and within-group (0.13, 0.24) change were observed in Dyrbye's RCT (46).

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These results are reflected in the individual study results *(Table 1)*. Barcons (44) did not detect any significant between-group differences for overall job satisfaction, whilst Margalit (50) demonstrated significant improvement in self-esteem between groups.

b. Organisational interventions (k=4)

One RCT and three NCBAs trialled organisational interventions. Means and standard deviations were not provided; therefore, we were not able to calculate effect sizes. Linzer et al. (RCT, n=166) demonstrated that workflow interventions, communication and overall quality improvements benefited satisfaction in the intervention group (49). Whilst Dunn et al. (NCBA, n=32), showed that quality improvement projects in the workplace showed significant improvement in quality work competence ratings but fluctuating satisfaction levels (2). Two smaller uncontrolled trials (n=13 in total) investigated the addition of clerical staff to the practice. Pozdnyakova et al. showed that the addition of clerical staff led to an improvement in satisfaction with the clinic workflow from 2/6 to all 6 GPs in a single practice, but did not report on any other measures of wellbeing (52). Similarly, Contratto et al. reported improved quality of life and professional balance for seven general medicine physicians in a mixed-methods approach (45).

c. Systemic interventions (k=2)

The introduction of a new contract with pay for performance scheme showed a significant improvement in job satisfaction (56) with an effect size of 0.44 between 2004 and 2005. Allen et al. used the same data and included a 2008 survey to look at satisfaction as a function of the exposure of GPs to the pay per performance scheme. Whilst job satisfaction declined again in 2008, the exposure to the scheme didn't affect satisfaction (41).

Risk of bias

The types of intervention and study settings precluded blinding for randomised controlled studies (no allocation concealment for waitlist control groups), and the outcome measures were participant reported throughout, and as such all studies were rated as high risk of bias by the Cochrane RoB2.

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DISCUSSION

Strengths, limitations and interpretation of evidence

We identified nineteen studies, half of which were published in the last four years, demonstrating an increased interest in the improvement of wellbeing and satisfaction of GPs. In comparison, a systematic review from 2016 looking at interventions to reduce burnout in physicians included fifteen RCTs and 37 cohort studies with 20 studies conducted before 2010 (57), suggesting burnout has been a research focus for longer. The considerable heterogeneity in the definition and measurement of constructs, study design, participant numbers, intervention types, intervention dose (ranging from six to 53 hours), follow-up periods, quality and reporting precluded a meta-analysis. It is challenging to draw robust conclusions regarding the (comparative) effectiveness of the different types of interventions reviewed.

Mindfulness interventions provided the most comprehensive and robust data with moderate to large effect sizes on mindfulness outcomes, and low to moderate effect sizes on compassion, resilience and empathy. Our results are supported by two reviews looking at the effects of mindfulness-based interventions on wellbeing (58, 59), more generally. Lomas et al. conducted a systematic review and meta-analysis and assessed 41 studies with approximately 2,100 participants. They found a wide range of self-reported outcomes (with both positive and negative measures of wellbeing). Reported within-group effect sizes for mindfulness, positive wellbeing (or life satisfaction), and compassion hovered around a moderate 0.5 mark, ES for empathy was 0.31; whilst for RCTs, the between-group ES for mindfulness, life satisfaction, and compassion were around 0.3 (58).

Scheepers et al. contributed a narrative review of 23 studies looking at mindfulness-based interventions for wellbeing in doctors of all ages and specialities. Review authors noted some caveats; considerable variation in type and dose-intensity of practice, sparse long-term outcome data, and methodological limitations. They cautioned that mindfulness practice involves time and dedication, which is not always feasible for busy health care professionals. In sum, the conclusions they drew are tentatively positive.

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In contrast to Lomas et al. our ES for mindfulness are higher between-group than within-group, which is somewhat unusual. This may be explained by a small study effect, and it may be driven by one study (42) with a decline in mindfulness over time for their control group.

We identified four studies evaluating coaching and experiential workshops for GPs, which showed low effect sizes for satisfaction measures and moderate ES for work-related morale and quality of work life. There does not appear to be much literature on coaching for health care professionals. One quasi-experimental study by Gardiner et al. looked at 'cognitive behavioural coaching' in rural Australian GPs and demonstrated a significant within-group reduction in distress levels for the coachees (60). Resilience training for a range of different physicians that had completed training was investigated in a recent systematic review. Four RCTs and five observational studies were included. The authors flag heterogenous study design and use of outcome measures, as well as quality issues with weak evidence for the interventions, whilst indicating potential for improvement of resilience (61).

We found four small-scale organisational interventions that suggested improved (job) satisfaction, as did one large-scale health policy intervention of performance-related pay in the UK. For burnout, a paucity of interventions trials delivered at organisational and systemic levels has been previously reported (62, 63), the authors suggest to actively design such trials. Similarly, Dyrbye et al., concluded that whilst useful, an individual intervention such as coaching is no replacement for organizational improvement (46). Shanafelt and colleagues have collated their vast research into burnout and put forward nine organisational strategies to address burnout and physician wellbeing through leadership (64). Despite calls for action, these avenues have not been adequately addressed or reported to date, at least not for GPs, and may warrant further exploration. Considering the time it takes to gather and report data, it is understandable that organizations might feel pressure to implement programs based on preliminary data.

Suggestions for future research

For peer review only

Based on our findings, we provide some suggestions which may be useful for future research into wellbeing and satisfaction for GPs. Stronger collaboration amongst researchers in this space may also lead to improved results.

a) Selection of outcomes & outcome measures

The reduction in burnout and stress is often equated with an improvement in wellbeing or satisfaction. We argue that the improvement of negative outcomes doesn't necessarily indicate a presence of satisfaction or wellbeing. This aligns with the dual continuum model of mental health / mental ill-health and flourishing / languishing (25-28). Good mental health isn't automatically linked to flourishing, nor is mental ill-health an indicator of languishing. Other authors have made similar statements (10, 24, 42, 58). We did not find a single study about flourishing in GPs.

Clearly defining the constructs 'wellbeing' and 'satisfaction', whilst utilising validated wellbeing and satisfaction measures, will enhance clarity, consistency and comparability of study design and reporting. Brady et al., who conducted a systematic review looking at the definition and measurement of 'physician wellness' stated that there needs to be consensus and clarity of definition, if we want to improve the quality and comparability of research in this space (65). Whilst this would improve the next phase of studies, the urgency in calls for actions may need to be balanced against the calls for consistency among studies.

b) Organisational and systemic interventions

When looking at known drivers of burnout (organisational culture, workplace conditions, lack of control and autonomy), it's not surprising that individual interventions are not as effective as desired (49, 62, 66-68). Hence, more combined approaches targeting both individuals and organisations have been proposed. The same might be true for wellbeing interventions.

A 2017 British meta-analysis contrasted different types of interventions for burnout on the individual doctor and on the systemic level, whereby systemic interventions appear more effective (69). Similarly, groups in the USA state that the approach must be combined and include organisational interventions (1, 70, 71), mostly focusing on time management, rostering, workflow management, staffing and use of information

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technology solutions. Overall, there is a scarcity of organisational interventions aimed at reducing burnout (72), and conclusions from the two meta-analyses of interventions to reduce burnout should be considered provisional. We endorse an intensified effort to explore organisational interventions to improve wellbeing and satisfaction.

c) Physical interventions

We didn't find any physical interventions (i.e. exercise, nutrition) geared towards improving GPs' wellbeing. Sparse research on exercise or diet interventions for doctors exists. A Pakistani cross-sectional survey revealed that 76% of nearly 1200 doctors, nurses and dentists did not exercise at all, and only one participant ate according to USDA dietary guidelines (73). While a US cross-sectional survey of 303 physicians found that less than 25% knew the American Heart Association (AHA) dietary recommendations, whilst around half knew and followed their physical activity guidelines (74). Two systematic reviews looked at exercise and burnout in the general population, one was inconclusive (75), the other stated that physical activity effectively reduces burnout (76). Both identified methodological issues and no long term follow-up. Seeing the paucity of data, this might provide an avenue for further investigation.

d) Quality and risk of bias

Areas for risk of bias are inherent in this type of research. However, measures can be taken to reduce bias for example by using active controls in randomized studies as suggested by other review authors (59), by consistently publishing study protocols a priori, and controlling for participant attributes, such as prior engagement in mindfulness practice. Ideally, the same rigourous approach should be applied to intervention studies for clincians, as to clinical interventions studies for patients.

CONCLUSION

Despite increasing interest in the improvement of wellbeing and satisfaction, there is remarkably little evidence beyond mindfulness interventions aimed at individual GPs. Few studies utilize validated measures of wellbeing and satisfaction, and favour burnout tools. Studies looking into organisational and systemic interventions remain sparse, and conclusions about their effectiveness may be permature.

For peer review only

In light of the COVID-19 pandemic and the added strain to primary care, programmes to support and research GP wellbeing should be prioritised by policymakers and governments worldwide.

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Registration

The protocol was submitted for registration on PROSPERO, due to high workload a registration is still pending.

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Competing interest statement

All authors have completed the Unified Competing Interest Form.

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Ian Hickie has declared financial relationships outside of the submitted work in the previous three years and other relationships or activities.

Authors' Contributions

DNA is the guarantor and corresponding author and attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted.

Authors contributed to the study conception and design (DNA, JKH, RE, IH, NG), the acquisition (DNA, AS, NG), analysis (DNA, AS, NG) and interpretation (DNA, AS, RE, IH, NG) and the drafting or critical revision of important intellectual content and final approval of the version to be published and agree to be accountable for all aspects

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of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Transparency declaration

Dr Diana Naehrig (the manuscript's guarantor) affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned and registered have been explained.

Data sharing statement

All included studies are published. We will consider sharing data upon reasonable request.

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For peer review only

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For peer review only

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Table 1. Evidence table

Author, Date, Study design	Partici-pants (n, mean age, % male)	Interven-tion (n, mean age)	Control (n, mean age)	Positive outcome measure	Other outcome measure	Baseline scores	Post-intervention score	Follow-up score (mean, SD, n)	Results - Summary
Allen 2017 Panel survey linked with QOF data 2004/5 (first year) and 2007/8 (fourth year)	GPs in the UK. 2004: n=1950, 47.0yrs, 66.2% male. 2005: n=2000, 47.9yrs, 63.6 % male. 2008: n=1986, 48.7yrs, 63.3% male.	Pay for perform-ance (P4P) scheme; Quality and Outcomes Framework (QOF) introduced in NHS in 2004	NA	Job satisfacti on (JSS WCW). Life satisfacti on	P4P exposure	Mean. 2004: Life satisfaction (4.649). Overall JS (4.567), Physical working conditions (4.862), Choose method of working (4.636), Colleagues (5.515), Recognition for good work (4.224), Responsibility (4.976), Remuneration (4.376), Opportunity to use abilities (4.787), Hours of work (3.914), Variety in job (5.011).	Time at 12 months. Mean. 2005: Life satisfaction (5.095). Overall JS (5.201), Physical working conditions (5.044), Choose method of working (4.892), Colleagues (5.599), Recognition for good work (4.726), Responsibility (5.406), Remuneration (5.387), Opportunity to use abilities (5.147), Hours of work (4.802), Variety in job (5.269).	Time at 48 months. Mean. 2008: Life satisfaction (5.008). Overall JS (4.728), Physical working conditions (5.129), Choose method of working (4.640), Colleagues (5.602), Recognition for good work (4.495), Responsibility (5.276), Remuneration (4.849), Opportunity to use abilities (5.074), Hours of work (4.205), Variety in job (5.276).	Between 2004 and 2005 overall job satisfaction increased (also see Whalley 2008), then fell from 2005 to 2008. There is a positive effect (though not statistically significant) between QOF/P4P income exposure and job satisfaction in 2005 (t-ratio 1.74), though not so in 2008 (t-ratio 0.14). The P4P exposure shows now significant effect on GPs job satisfaction.
Amutio 2015 RCT	Physic-ians (approx 70% primary care) in public or private practice in Spain (n=42, 47.3 yrs, 42.9%)	8 week MBSR programme (group course 8x2.5h/wk, 1x8h retreat plus homework) , 10 month mainten-ance period (group course 10x2.5h/mt plus homework) (n=21)	WLC (n=21)	Mindful-ness (FFMQ), Relax-ation (SRSI-3)	Heart rate	Mean and SD. INTERVENTION (n=21) FFMQ Total (3.34, 0.44), FFMQ Observing (3.33, 0.60), FFMQ Describing (3.58, 0.72), FFMQ act with awareness (3.16, 0.87), FFMQ Nonjudging (3.42, 0.64), FFMQ Nonreactivity (3.17, 0.51). SRSI relaxation (2.54, 0.53), SRSI positive energy (3.09, 0.64), SRSI mindfulness (3.74, 0.89), SRSI Transcedence (2.67, 0.82). CONTROL (n=21) FFMQ Total (3.51, 0.25), FFMQ Observing (3.02, 0.88), FFMQ Describing (3.88, 0.53), FFMQ Act with awareness (3.93, 0.70), FFMQ Nonjudging (4.19, 0.52), FFMQ Nonreactivity (3.23, 0.57). SRSI relaxation (2.52, 0.5), SRSI positive energy (3.01, 0.62), SRSI mindfulness (4.29, 0.92), SRSI	Time at 8 weeks. Mean and SD. INTERVENTION (n=21) FFMQ Total (3.71, 0.51), FFMQ Observing (3.98, 0.64), FFMQ Describing (3.83, 0.62), FFMQ act with awareness (3.48, 0.65), FFMQ Nonjudging (3.78, 0.66), FFMQ Nonreactivity (3.46, 0.63). SRSI relaxation (3.08, 0.61), SRSI positive energy (3.80, 0.82), SRSI mindfulness (4.45, 0.71), SRSI Transcedence (3.27, 1.02). CONTROL (n=21) FFMQ Total (3.34, 0.33), FFMQ Observing (2.83, 0.98), FFMQ Describing (3.82, 0.58), FFMQ Act with awareness (3.91, 0.61), FFMQ Nonjudging (4.16, 0.52), FFMQ Nonreactivity (3.26, 0.67). SRSI relaxation (2.60, 0.51), SRSI positive energy (3.01, 0.6), SRSI mindfulness	Time at 12 months. Mean and SD. INTERVENTION FFMQ Total (3.85, 0.49), FFMQ Observing (4.09, 0.62), FFMQ Describing (4.01, 0.58), FFMQ act with awareness (3.55, 0.69), FFMQ Nonjudging (3.96, 0.66), FFMQ Nonreactivity (3.58, 0.55). SRSI relaxation (3.16, 0.78), SRSI positive energy (3.99, 0.81), SRSI mindfulness (4.60, 1.01), SRSI Transcedence (3.65, 1.36).	The MBSR programme (statistically) significantly improves mindfulness and relaxation measures (particularly positive energy and transcendence) at 8 weeks after baseline. Sustained and even improved longterm effects at 12 months follow-up are shown.

						Transcedence (2.64, 0.93).	(4.24, 0.95), SRSI Transcedence (2.40, 0.92).		
Asuero 2014 RCT (2010 - 2012)	Primary health care profes-sionals (41 physician s) in Spain, public system (n=68 total, 47 yrs, 8% male)	8 week MB psycho-educational program: Didactic material, mindful-ness meditation, narrative/a ppreciative enquiry, discussion. Group course 8x2.5h/wk, 1x8h retreat plus homework. (n=43 total, 23 physicians, 48.8yrs)	WLC (n=25 total, 18 physi-cians, 46.9yr)	Mindfuln ess (FFMQ), Empathy (JSPE), also self-report on energy, wellbeing and activity	Burnout (MBI), mood disturban ce (POMS), intervent ion evaluatio n	Mean and SD. INTERVENTION (n=43). FFMQ Total (129.6, 22.2), FFMQ Observing (26.1, 8.6), FFMQ Describing (28.2, 5.7), FFMQ act with awareness (25.3, 5.9), FFMQ Nonjudging (27.1, 8.0), FFMQ Nonreactivity (21.9, 4.3). JSPE Total (119.5, 13.1), JSPE Perspective taking (54.8, 6.5), JSPE Compassionate care (47.2, 5.1), JSPE standing in pts shoes (13.1, 1.8). CONTROL (n=25). FFMQ Total (120.5, 14.4), FFMQ Observing (24.5, 4.3), FFMQ Describing (26.4, 5.2), FFMQ Act with awareness (23.6, 6.8), FFMQ Nonjudging (26.3, 6.5), FFMQ Nonreactivity (19.8, 2.9). JSPE Total (120.8, 10.1), JSPE Perspective taking (59.8, 7.0), JSPE Compassionate care (47.4, 4.9), JSPE standing in pts shoes (13.6, 0.9).	Time at 8 weeks. Mean and SD. INTERVENTION (n=43) FFMQ Total (141.6, 16.5), FFMQ Observing (30.4, 5.1), FFMQ Describing (28.9, 5.3), FFMQ act with awareness (27.4, 4.7), FFMQ Nonjudging (30.6, 6.2), FFMQ Nonreactivity (24.1, 3.0). JSPE Total (123.0, 9.2), JSPE Perspective taking (56.1, 5.3), JSPE Compassionate care (49.0, 3.9), JSPE standing in pts shoes (13.5, 1.1). CONTROL (n=25) FFMQ Total (121.5, 16.0), FFMQ Observing (24.1, 4.5), FFMQ Describing (26.5, 5.5), FFMQ Act with awareness (23.0, 5.9), FFMQ Nonjudging (27.4, 6.8), FFMQ Nonreactivity (20.2, 3.5). JSPE Total (119.0, 10.7), JSPE Perspective taking (59.6, 6.3), JSPE Compassionate care (46.9, 4.6), JSPE standing in pts shoes (12.5, 3.0).	NA	For mindfulness total there is significant improvement for the intervention (moderate change). For empathy total there is a non-significant increase from pre to post measure in the intervention group, only an increase in compassionate care was statistically significant. Overall mindfulness and empathy (compassionate care) were improved by the programme.

Barcons 2019 CBA, mixed methods, 2016 - 2017	Primary health care profess-ionals in Spain, public system (n=38).	MTP and IBST group programme plus routine programme . 9 x 1hr per week, (6h Training on psychology, 3h on psychiatry and 1h social work). (n=20)	Rout-ine mental -health sup-port pro-gram for prim-ary care (n=18)	Job Satisfacti on (FR-JS)	Burnout (MBI), Brief Psychiatri c Rating Scale (BPRS), Psychoph armacolo gy use, opinions about mental illness, administr ative and health care indicator s	Median and IQR. INTERVENTION (n=20) FR satisfaction at work (15, 13, 16.25), FR work tension (12, 11, 14), FR Professional competence (5, 4, 6), FR work pressure (12, 10, 12.25), FR professional promotion (12, 9, 13), FR relationship superiors (4, 4, 6.5), FR relationship peers (6, 5.75, 7), FR extrinsic status (5.5, 4 ,6), FR monotony (4.5, 4, 6), FR total (75, 72, 77.5). BPRS total (23.50, 22, 24.25). CONTROL (n=18), FR satisfaction at work (14, 10.5, 16), FR work tension (13.5, 12.25, 16.75), FR Professional competence (4.5, 3, 6), FR work pressure (12, 12, 13), FR professional promotion (11, 10, 13.75), FR relationship superiors (6, 4, 6.75), FR relationship peers (5, 5, 6), FR extrinsic status (5, 5, 6), FR monotony (5, 4, 5.75), FR total (76, 73, 80.5).	Time at 10 months. Median and IQR. INTERVENTION (n=20) FR satisfaction at work (16, 15, 17), FR work tension (13, 11, 16), FR Professional competence (5, 4.5, 6), FR work pressure (12, 10, 12.50), FR professional promotion (12, 10, 13), FR relationship superiors (4, 4, 6.5), FR relationship peers (5, 5, 6), FR extrinsic status (5, 5, 6.5), FR monotony (5, 4, 5), FR total (76, 73, 83). BPRS total (20.50, 19, 22). CONTROL (n=18), FR satisfaction at work (12, 9, 16), FR work tension (15, 1, 16), FR Professional competence (5, 4, 6), FR work pressure (11, 11, 12), FR professional promotion (12, 11, 15), FR relationship superiors (6, 4, 8), FR relationship peers 6, 5, 7), FR extrinsic status (6, 4, 6), FR monotony (4, 3, 5), FR total (77, 75, 78).	NA	No statistically significant changes in job satisfaction (FR). For wellbeing, a statistically significant decrease was seen from pre to post measure in the intervention group, as the BPRS measures negative constructs, this is interpreted as an improvement of wellbeing.
Contratto 2017 NCBA, mixed methods	General medicine physician s in the USA, working part-time (n=7)	1 clerical staff was hired in a GP practice to enter orders (n=7).	NA	Personal-professio nal balance, physician satisfacti on defined as QoL.	Burnout (MBI), physician product-ivity, EMR documen tation. Qual inter-views.	QoL: 14% bad, 14% neutral, 71% good. Balance: 14% Dissatisfied, 43% neutral, 43% good.	Time at 4 months. QoL: 0% bad, 0% neutral, 100% good. Balance: 0% Dissatisfied, 29% neutral, 71% good.		QoI, personal balance, burnout improved. Productivity increased (work relative value unit) per session increased. Reports of feeling more supported, focused on pt care, less stress, less fatigue

Dunn 2007 NCBA, 2000-2005	Primary care in urban USA (n=32, 25 internists 6 family medicine 1 NP)	Organisat-ional inter-vention programme to improve physician wellbeing (quality improve-ment project, participant data-guided interv-ention) (n= 22 to 32 range over the inter-vention period).	NA	Physician satisfacti on (ACP/ASI M)	Burnout (MBI), Quality Work Compete nce (QWC)	2001: 55% of physicians were somewhat or very satisfied with their practice	Time at 24 months. 2003: 84% were somewhat or very satisfied with their practice	Time at 24 months. 2005: 74% were somewhat or very satisfied with their practice.	Intervention ongoing from 2000 onward. Emotional and work-related exhaustion decreased significantly over the study period. QWC measures of organizational health significantly improved initially and remained acceptable and stable during the rest of the study. Satisfaction fluctuated throughout.
Dyrbye 2019 RCT, 2017 - 2018	Physician s in USA (67 generalis ts, 21 subspec-ialists). (n=88, 45.5% male).	6 coaching sessions (3.5h) facilitated by a profess-sional coach over 5mts (n=44).	WLC (n=44)	Resilien-ce (CD-RISC), Global Job Satisfact-ion-12 (subscale of PJSS), work engage-ment (UWES), empower ment at work scale.	Burnout (MBI).	Means and SD. INTERVENTION (n=44), Resilience (31.0, 6.3), Job satisfaction (43.4, 10.7), UWES vigor (3.9, 1.0), UWES Dedication (4.5, 1.1), UWES Absorption (4.2, 1.0), Empowerment at work (55.5, 11.9). CONTROL (n=44), Resilience (30.6, 5.7), Job satisfaction (42.8, 10.6), UWES vigor (4.0, 1.2), UWES Dedication (4.6, 1.0), UWES Absorption (4.1, 1.1), Empowerment at work (57.3, 14.0).	Time at 5 months. Means. INTERVENTION (n=44), Resilience (32.3), Job satisfaction (44.4), UWES vigor (4.1), UWES Dedication (4.6), UWES Absorption (4.1), Empowerment at work (58.2). CONTROL (n=44), Resilience (31.2), Job satisfaction (43.2), UWES vigor (4.2), UWES Dedication (4.7), UWES Absorption (4.2), Empowerment at work (60.3).		Statistically significant improvement for resilience from pre to post intervention, no change in job satisfaction. Burnout, emotional exhaustion decreased. QoL improved. No statistically significant differences in depersonalization, engagement, or meaning in work.
Fortney 2013 nCBA	Family medicine practit-ioners in USA (n=30, 87% family medicine physic-ians)	Shortened MBSR course, 18h total, 14h weekend, 2x 2h plus homework (n=30).	NA	Compas-sion (SCBC), Resili-ence (RS-14).	Burnout (MBI), Depressi on & Anxiety (DASS), Perceive d Stress scale (PSS),	Mean score and Confidence Interval. n=30. Resilience RS-14 (79.9, CI 75.2-84.6), Compassion SCBC (27.6, CI 25.9-29.3).	Time at 4 weeks. Mean score and Confidence Interval. n=28. Resilience RS-14 (82, CI 77.1-86.8), Compassion SCBC (27.4, CI 25.6-29.1).	Time at 9 months. Mean score and Confidence Interval at 9mts. n=23. Resilience RS-14 (81.4, CI 76.2-86.6), Compassion SCBC (28.3, CI 26.5-30.1).	No significant improvement in compassion and resilience over time. Participants had improvements compared with baseline at all follow-up time points for Burnout, depression, anxiety and perceived stress.

Gardiner 2004 CBA	GPs in metro-politan Australia (n=110)	Cognitive behavioural stress manage-ment course, 1x 3h, 15h over 5 wks. (n=85).	Control group had other devel-opmental courses. (n=25)	Work related morale measure (WRM-7), Quality of work life (QoWL-6)	Work related distress measure (WRD-7), General psychological distress (GHQ-12), coping styles.	Means and SD. Total measure without sub-measures given.INTERVENTION (n=86), Work related morale total 31.83 (6.75). QOWL total 25.32 (6.64). CONTROL (n=24), Work related morale total 31.42 (6.19). QoWL total 23.16 (5.86).	Time at 4 weeks. Means and SD. INTERVENTION (n=77), Work-related morale total 34.62 (6.11). QoWL total 28.24 (6.35). CONTROL (n=19), Work related morale total 32.21 (6.73). QoWL total 24.32 (6.36).	Time at 12 weeks. Means and SD. INTERVENTION (n=62), Work related morale total 35.70 (6.01).	Overall the intervention group showed higher scores post intervention than the control group (positive trend). However, no statistically significant change seen for work related morale or QoWL for the intervention group. Only physicological distress significantly lower for intervention group. When looking at those GPs that scored low for morale, there was a 56% reduction pre to post intervention, compared with a 29% reduction in the control group.
Krasner 2009 NCBA	Primary care physici-ans in the USA (n=70, 54% male)	CME programme : Didactic material, mindful-ness meditation, narrative/appreciative enquiry, discussion. 28h total. 8 x 2.5hr per week, 1x 7h, 10-month mainten-ance phase (2.5 h/mo). (n=70)	NA	Mindfulness (Baer), Empathy (JSPE)	Burnout (MBI), Mood (POMS), Big 5 personality factors, Physician belief (PBS)	Mean and CI. Baer mindfulness total (45.2, 95% CI 43.3-47.1, n=60), Baer mindfulness observe (25.6, 95% CI 24.4-26.8, n=60), Baer mindfulness nonreact (19.7, CI 95% 18.7-20.7, n=60). JSPE total (116.2, 95% CI 114.2-118.9, n=60), JSPE compassionate care (48.6, 95% CI 47.5-49.7, n=60), JSPE perspective taking (57.1, 95% CI 55.6 to 58.6, n=60), JSPE standing in pts shoes (10.8, 95% CI 10.4 to 11.5, n=60).	Time at 8 weeks. Mean and CI. Baer mindfulness total (52.9, 95% CI 51 to 54.8, n=59), Baer mindfulness observe (30.6, 95% CI 29.4 to 31.8, n=59), Baer mindfulness nonreact (22.9, CI 95% 21.8 to 23.9, n=59). JSPE total (120.6, 95% CI 118.2 to 123.0, n=59), JSPE compassionate care (49.8, 95% CI 48.7 to 50.9, n=59), JSPE perspective taking (59.1, 95% CI 57.6 to 60.6, n=59), JSPE standing in pts shoes (11.7, 95% CI 11.1 to 12.2, n=59).	Time at 12 months. Mean and CI. Baer mindfulness total (55, 95% CI 53.0 to 56.9, n=56), Baer mindfulness observe (31.1, 95% CI 29.8 to 32.3, n=56), Baer mindfulness nonreact (23.9, CI 95% 22.9 to 24.9, n=56). JSPE total (121.4, 95% CI 119.0 to 123.8, n=56), JSPE compassionate care (50.4, 95% CI 49.3 to 51.5, n=56), JSPE perspective taking (59.7, 95% CI 58.2 to 61.2, n=56), JSPE standing in pts shoes (11.4, 95% CI 10.9 to 11.9, n=56).	Baseline scores and follow-up scores at 15 months are not reported for the purpose of this SR. Over time, all measures for mindfulness, burnout, physician belief, mood and, personality improved, the largest effect size was observed for mindfulness at 15 mts.
Linzer 2015 RCT	Primary care clinicians in the USA (n=166, > 80% physic-ians, 47.3 yrs, 48% male)	Interven-tions to improve communica-tion; workflow, quality improve-ment (QI). n=83	WLC n=83	Work Control, Satisfaction (from PWS and MEMO).	Stress, burnout, chaos, intent to leave, varia	Number given in %. INTERVENTION (n=83) High work control (score >2) 96%, High satisfaction (>3) 38.5%. CONTROL (n=83) High work control 13.2%. High satisfaction (>3) 51.8%.	Time at 12 to 18months. Number given in %. INTERVENTION (n=67) High work control (score >2) 4.6%, High satisfaction (>3) 40.0%. CONTROL (n=72) High work control 11.4%, High satisfaction (>3) 45.7%.		Satisfaction improved with workflow interventions, targeted QI projects, communication.

Margalit 2005 RCT	GPs in Israel (n=44)	CPD teaching programs (n=22 for interactive program)	CPD teach-ing progr-am (n=22 didac-tic meth-od)	Attitude to patient care, self-esteem	Knowledge, intention, attitude, burnout	Mean, SD. Self-esteem (72.1, 14.5), Attitude to patient care (34.5, 12.2) (n=44)	Time at 6 months. Mean, SD. Self-esteem (76.5, 12.9), Attitude to patient care (36.5, 19.9) (n=44)	NA	Significant improvement on self-esteem post intervention. The interactive teaching approach improved self-esteem more than the didactic teaching did. No improvement on attitude to patient care.
Montero-Martin 2018 NCBA	GPs in Spain (n=290, 49 yrs, 22.5% males)	Brief blended web-based mindful-ness interven-tion,1x 4h face to face, 8 online sessions (2 weekly sessions over 4 wks) (n=58).	NA	Positive affect (PANAS), awareness (MAAS), resilience (CD-RISC)	Negative affect (PANAS), Burnout subtypes(BCSQ)	Mean, SD. 1 session/wk (n=28): PANAS-pos (32.19, 6.72). MAAS (64, 12.07). CD-RISC (38.96, 8.96). 2 sessions/wk (n=30). PANAS-pos (32.03, 6.38). MAAS (61.77, 13.41). CD-RISC (38.80, 8.58).	Time at 4 weeks. Mean, SD. 1 session/wk (n=28): PANAS-pos (33.44, 5.42). MAAS (66.67, 10.88). CD-RISC (40.19, 5.17). 2 sessions/wk (n=30). PANAS-pos (35.00, 4.91). MAAS (66.37, 11.03). CD-RISC (41.28, 4.32).		Benefits in pos PANAS and MAAS for two or more weekly meditation session. No benefits for one weekly practice. While face to face attendence was good, very high attrition rate for online component.
Pozdnya-kova 2018 NCBA, 2007	Academic general internal medicine clinic (n=6 faculty, n=325 patients).	Clinic sessions with and without a scribe.	NA	Physician workplace satisfaction	Burnout, Time spent on EHR documentation, patient satisfaction with doctor–patient relationship, attitudes towards scribes.	n=6. Number of responses 'agree' or 'strongly agree'. Satisfied with clinic workflow 2/6 (33%). Calm atmosphere in work area 0/6 (0%). Satisfied with quality of patient interactions 5/6 (83%). Satisfied with quality of communication with patient 4/6 (67%).	Time at 3 months. n=6. Number of responses 'agree' or 'strongly agree'. Satisfied with clinic workflow 6/6 (100%). Calm atmosphere in work area 2/6 (33%). Satisfied with quality of patient interactions 6/6 (100%). Satisfied with quality of communication with patient 5/6 (83%).		Of six physicians, 100% were satisfied with clinic workflow post-pilot (vs. 33% pre-pilot), and 83% were satisfied with EHR use post-pilot (vs. 17% pre- pilot). Physician burnout was low at baseline and did not change post-pilot. Mean time spent on post-clinic EHR documentation decreased from 1.65 to 0.76 h per clinic session (p = 0.02).
Rees 2020 NCBA, mixed methods	Rural medical practitioners in Australia (57% GPs). (n=13 total, n=4 qual research, n=7 quant research, 40 yrs, 76.9% male)	MSCR, Mindful Self-Care and Resiliency program (7h face-to-face session) and 3x 1h video-conference (follow up sessions)	NA	Wellbeing (WHO-5), Positive affect (PANAS).	Burnout measure (short version 10-item), psychological strain (GHQ-12)	n=7. Mean. Wellbeing (61.1). Positive affect not reported.	Time at 4 weeks. n=7. Mean. Wellbeing (71.4). Positive affect not reported.		For the WHO-5 wellbeing scale there was no change pre-post for one doctor, deterioration for two doctors with no reliable change, Improvement for four doctors with no reliable change.

Schroeder 2018 RCT, 2014 - 2015	Primary care physici-ans in the USA (n=33, 42.76 yrs, 27% male)	Mindful Medicine Curriculum = modified version of MBSR, 1x 13h & 2x 2h (n=16)	WLC (n=17)	Mindful Attention Awarene ss (MAAS), Resilienc e (BRS), Compassi on (SCBC),	Stress (PPS), Burnout (MBI), Meditati on Practice (MPQ)	Mean, SD. INTERVENTION (n=16) MAAS (3.42, 0.96), BRS (21.62, 4.45), SCBCS (26.31, 4.51). CONTROL (n=17) MAAS (3.32, 0.76), BRS (18.70, 5.13), SCBCS (27.00, 4.97).	Time at 4 weeks. Mean, SD. INTERVENTION (n=15) MAAS (3.62, 0.89), BRS (22.33, 4.74), SCBCS (27.66, 3.22). CONTROL (n=14) MAAS (3.08, 0.76), BRS (19.42, 4.21), SCBCS (26.07, 4.73).	Time at 3 monthrs. Mean, SD. INTERVENTION (n=13) MAAS (4.04, 1.02), BRS (24.15, 5.47), SCBCS (27.84, 4.09). CONTROL (n=13) MAAS (3.18, 0.58), BRS (18.82, 5.32), SCBCS (25.07, 5.85).	Participants in the MMC group reported significant improvements over time for MAAS (also PSS and MBI), whereas in the control group, there were no improvements. There were no significant improvement for resilience or compassion.
Van Vietmar-schen 2018 NCBA, mixed methods, 2015 - 2016	Primary care physic-ians in the Nether-lands (n=54, 87% GPs, 40 yrs, 22% male)	Adapted MBSR programme , weekly group sessions for 8 weeks, 26 hours total.	NA	Self-Compassi on (SCS), Self-reflection (GRAS).	Cohen Perceive d Stress Scale (PSS)	Mean, SD. SCS (2.9, 0.7, n=50), GRAS (87.6, 7.7, n=44).	Time at 8 weeks. Mean, SD. SCS (3.4, 0.6, n=50), GRAS (90.9, 6.7, n=44).	Time at 6 months. Mean, SD. SCS (3.7, 0.7, n=21), GRAS (90.2, 10.9, n=17).	Significant improvement of self-compassion and self-reflection. Six months after PSS and SCS were still improved. PSS significantly reduced. Qual: awareness, acceptance, peacefulness and openness improved through intervention.
Duerweij 2016 CBA, mixed methods	General practiti-oner trainers in the Nether-lands (n=50, 54.9 yrs, 66% male)	MBSR training 8x 2.5h, 1x 8h retreat (n=30)	WLC (n=20)	Empathy (JSPE-20), Mindfuln ess (FFMQ-39)	Work engagem ent, Burnout (UBOS-C)	Mean, SE. INTERVENTION (n=43) JSPE (117.4, 1.53), FFMQ Total (136.21, 2.23), FFMQ Observing (26.36, 0.69), FFMQ Describing (28.26, 0.9), FFMQ Acting with awareness (27.12, 0.71), FFMQ Non-judging (31.16, 0.81), FFMQ Non-reacting (23.34, 0.57). CONTROL (n=20) JSPE (116.18, 1.92). FFMQ Total (135.48, 2.65), FFMQ Observing (25.85, 0.82), FFMQ Describing (28.33, 0.9), FFMQ Acting with awareness (28, 0.9), FFMQ Non-judging (30.49, 0.98), FFMQ Non-reacting (23.12, 0.7).	Time at 8 weeks. Mean, SE. INTERVENTION (n=43) JSPE (119.35, 1.49), FFMQ Total (143.08, 2.19), FFMQ Observing (28.4, 0.68), FFMQ Describing (29.77, 0.71), FFMQ Acting with awareness (28.1, 0.69), FFMQ Non-judging (32.36, 0.79), FFMQ Non-reacting (24.47, 0.56, 0.57). CONTROL (n=20) JSPE (117.93, 1.98). FFMQ Total (135.45, 2.67), FFMQ Observing (26.33, 0.83), FFMQ Describing (28.2, 0.91), FFMQ Acting with awareness (27.41, .91), FFMQ Non-judging (30.32, 0.99), FFMQ Non-reacting (23.49, 0.7).		Mindfulness skills increased significantly in the MBSR group. Empathy remained the same. The qualitative data indicated that the MBSR course increased their wellbeing and compassion towards themselves and others, including their patients.

Whalley 2008 Panel survey, 2004, 2005	GPs in the UK. 2004: n=2105, mean age 46.9yrs, 66% male. 2005: n=1349, mean age 48.6yrs, 65 % male.	New introduct-ion of pay for perform-ance system happened in 2004 (after the 2004 survey)	NA	JSS WCW	Job pressure, Job design and time pressures	Mean, SD. 2004: JSS total (4.58, 1.39, n=2081)	Time at one year. Mean, SD. 2005: JSS total (5.17, 1.28, n=1345)		Statistically significant improvement in job satisfaction. Job pressure and work hours significantly declined. Most GPs reported that the new contract had increased their income (88%), but decreased their professional autonomy (71%), and increased their administrative (94%) and clinical (86%) workloads.

Key: RCT Randomised Controlled Trial; CBA Controlled Before and After Trial; NCBA Non-Controlled Before and After Trial; QOF Quality and Outcomes Framework; P4P Pay for Performance; NP Nurse Practitioner; WCW Warr Cook Wall; JS Job satisfaction; JSS Job Satisfaction Scale; MBI Maslach Burnout Inventory; JSPE Jefferson Scale of Physician Empathy); POMS Profile Of Mood States; FFMQ Five Facet Mindfulness Questionnaire; SRSI Smith Relaxation States Inventory; NA Not applicable; PBS Physician Belief Scale; SCS Self Compassion Scale (Neff); MPQ Meditation Practice Questionnaire, FR-JS Font Roja Job Satisfaction Questionnaire; BPRS Brief Psychiatric Rating Scale; IQR Inter Quartile Range; ACP College of Physicians; ASIM American Society of Internal Medicine; DASS Depression and Anxiety Scale; PJSS Physician Job Satisfaction Scale (3 dimensions JS, career satisfaction, and specialty satisfaction), 12-item Global Job Satisfaction subscale used; UWES Utrecht Work Engagement Score; CD-RISC Connor-Davidson Resilience Scale; PANAS Positive And Negative Affect Scale, MAAS Mindful attention awareness scale; GHQ General Health Questionnaire; AWS Areas of Work Life Scale (positive measures include control, reward, community, values); BRS Brief Resilience Scale; SCBCS Santa Clara Brief Compassion Scale (an abbreviation of the Sprecher and Fehr's Compassionate Love Scale); MPQ Meditation Practice Questionnaire; UBOS-C Utrecht Burnout Scale for Contactual Occupations (this is the validated Dutch version of the Maslach Burnout Inventory); GRAS Groningen Reflection Ability Scale; EHR Electronic Health Record; MTP Multimodal training programme; IBST integrated brief systemic therapy, PPS Perceived Stress Scale.

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Table 2. Overview of included studies

Intervention level	Intervention type	Study ID (Author, Date)	Study design	Outcome Measure	Within group effect sizes (experiment)	Between group effect sizes (exp vs control group)
Intervention targeted at the individual	Mindfulness	Amutio 2015	RCT	FFMQ	0.78	1.5
		Asuero 2014	RCT	FFMQ, JSPE	0.64, 0.31	0.57, 0.44
		Schroeder 2018	RCT	MAAS, BRS, SCBC	0.63, 0.51, 0.35	0.88, 0.61, 0.73
		Verweij 2016	CBA	FFMQ, JSPE	0.47, 0.20	0.5, 0.02
		Fortney 2013	NCBA	SCBC, RS-14	-0.04, 0.17	-
		Montero-Marin 2018	NCBA	PANAS, MAAS, CD-RISC	0.52, 0.37, 0.37	-
		Rees 2020	NCBA	PANAS, WHO-5	-, 0.52	-
		Van Wietmarschen 2018	NCBA	SCS, GRAS	0.77, 0.46	-
		Krasner 2009	NCBA	FFMQ-2 (BAER), JSPE	1.05, 0.44	-
	Educational training / experiential workshop	Barcons 2019	CBA	FR-JS	-	-
		Gardiner 2004	CBA	WRM-7, QoWL-6	0.43, 0.45	0.3, 0.27
		Margalit 2005	RCT	Self-esteem	-	-
	Coaching	Dyrbye 2019	RCT	CD-RISC, PJSS	0.24, 0.10	0.13, 0.06
Intervention at organisational level	Clerical support / Scribes	Contratto 2017	NCBA	Professional balance, physician satisfaction (QoL)	-	-
		Pozdnyakova 2018	NCBA	Physician workplace satisfaction	-	-
	Organisational improvement programme / QA	Dunn 2007	NCBA	Physician satisfaction	-	-
		Linzer 2015	RCT	Work control, satisfaction	-	-
Intervention at systemic level / policy	Introduction of pay for performance scheme	Allen 2017	NCBA (naturalistic)	Job satisfaction (JSS WCW), Life satisfaction	-	-
		Whalley 2008	NCBA (naturalistic)	JSS WCW	0.44	-

Key: Study design: RCT Randomised Controlled Trial; CBA Controlled Before and After trial; NCBA Non-Controlled Before and After trial.
Outcome measures: JSS WCW Warr Cook Wall Job satisfaction Scale; JSPE Jefferson Scale of Physician Empathy; FFMQ Five Facet Mindfulness Questionnaire; SRSI Smith Relaxation States Inventory; SCS Self Compassion Scale (Neff); FR-JS Font Roja Job Satisfaction Questionnaire; SCBC Santa Clara brief compassion scale (an abbreviation of the Sprecher and Fehr's Compassionate Love Scale); PJSS Physician Job Satisfaction Scale (3 dimensions JS, career satisfaction, and specialty satisfaction), 12-item Global Job Satisfaction subscale used; CD-RISC Connor-Davidson Resilience Scale; PANAS Positive And Negative Affect Scale, MAAS Mindful attention awareness scale; BRS Brief Resilience Scale; GRAS Groningen Reflection Ability Scale; RS-14 Resilience, WHO-5 Wellbeing Index, WRM Work Related Morale Measure. QoWL-6 Quality of Work Life.

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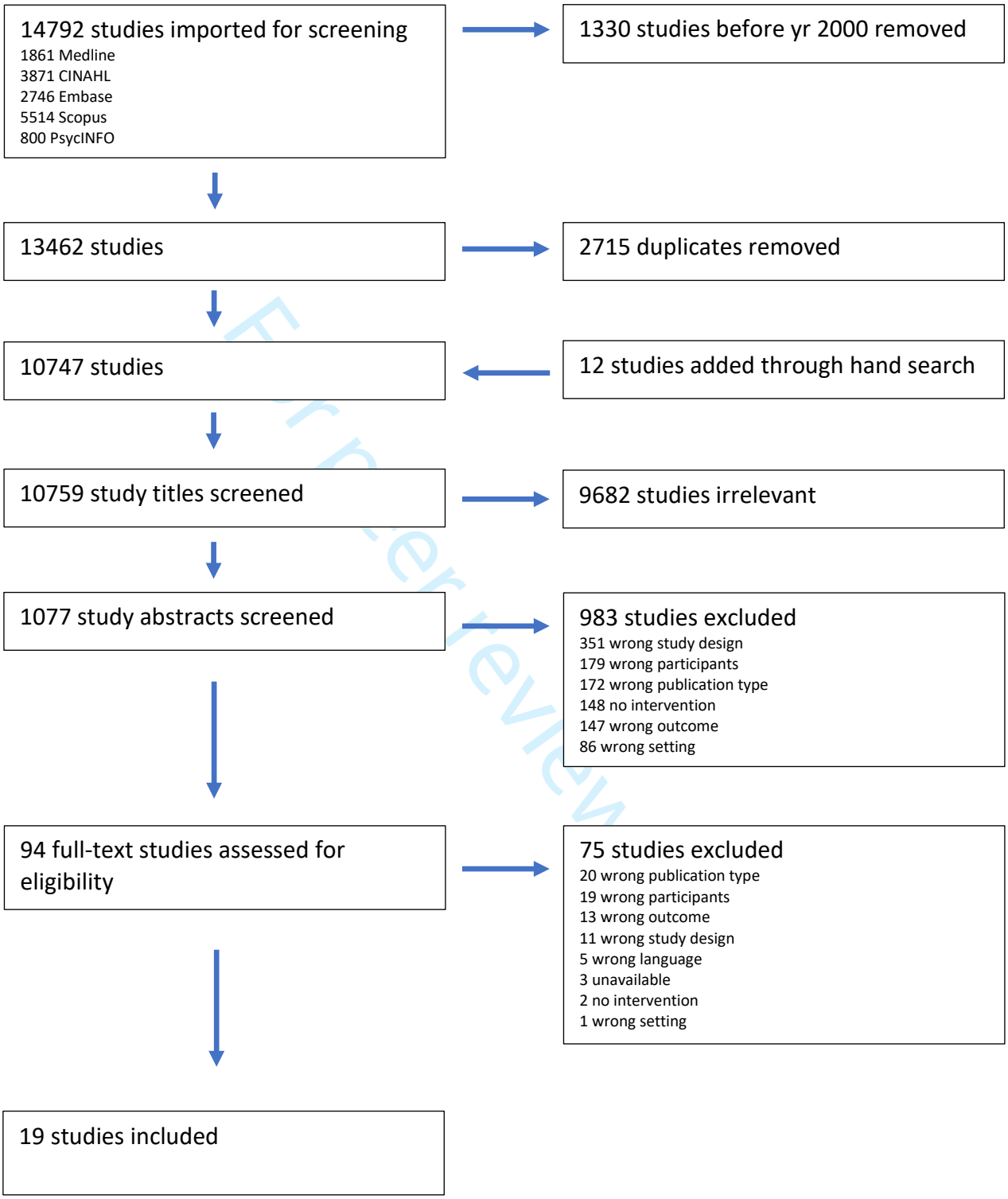
Table 3. Overview of within group and between group Effect Sizes (ES) for several positive outcomes of mindfulness interventions

	Outcome	Mindfulness	Resilience	Compassion	Empathy
Mindfulness interventions (k=9)	Within group ES (k=9)	0.37 – 1.05	0.17 – 0.51	-0.04 - 0.77	0.2 – 0.44
	Between group ES (k=4)	0.5 – 1.5	Only one study 0.61	Only one study 0.73	0.02 - 0.44

There were 9 studies that trialled mindfulness interventions (k=9). These studies utilized a range of different outcomes measures and included tools for assessing mindfulness, resilience, compassion and empathy. Within group effect sizes (ES) are shown, comparing before and after measures for the interventi

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Figure 1. PRISMA diagram



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Supplements

- Supplement 1. Medline OVID search strategy
- Supplement 2. Formulae used for Effect Size calculations in Excel for within and between group ES.
- Supplement 3. Effect sizes for outcome measures and sub-measures
- Supplement 4. Outcome measures utilized in the included studies.

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Supplement 1. Medline OVID search strategy

1. general practitioners/
2. "general practitioner*".mp.
3. (GP or GPs).mp.
4. exp General Practice/
5. Family Practice/ or Primary Health Care/
6. "family practitioner*".mp.
7. "primary care practitioner*".mp.
8. physicians, family/ or physicians, primary care/
9. "family medicine".mp.
10. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9
11. ("psychological well being" or "psychological wellbeing").mp.
12. (wellbeing or "well being").mp.
13. ("cognitive well being" or "cognitive wellbeing").mp.
14. flourishing.mp.
15. ((job or work) adj3 (satisf* or engage* or motivat*)).mp.
16. life satisfaction*.mp.
17. Job Satisfaction/
18. resilien*.mp.
19. Resilience, Psychological/
20. burnout.mp.
21. Burnout, Professional/
22. Compassion fatigue/
23. compassion fatigue.mp.
24. joy.mp.
25. "joy in practice".mp.
26. 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25
27. Motivational interviewing/
28. Mindfulness/
29. Mediation/

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30. mindful*.mp.

31. meditat*.mp.

32. Self-Care/

33. ("self care" or "self-care").mp.

34. Motivation/

35. Professional autonomy/

36. ("self determination" or "self-determination").mp.

37. exp Randomised Controlled Trial/ or Clinical Trial/

38. trial.mp.

39. Early Medical Intervention/ or intervention.mp.

40. Counseling/

41. Course?ling.mp.

42. Mentoring/

43. mentor*.mp.

44. Education/

45. education.mp.

46. workshop*.mp.

47. training.mp.

48. coaching.mp.

49. facilitat*.mp.

50. 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49

51. 10 and 26 and 50

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Supplement 2. Formulae used for Effect Size calculations in Excel for within and between group ES.

Within-group:

Mean_change_WG = Mean_Exp_post – Mean_Exp_pre

SD* pooled _WG = SQRT(((N_Exp_post-1) * SD_Exp_post^2) + (N_Exp_pre-1) * SD_Exp_pre^2)) / (N_Exp_post + N_Exp_pre – 2))

ES_WG = Mean_change_WG / SD* pooled_WG

Between-group:

Mean_change_BG = (Mean_Exp_post – Mean_Exp_pre) – (Mean_Ctrl_post – Mean_Ctrl_pre)

SD* pooled_BG = SQRT(((N_Ctrl_pre-1) * SD_Ctrl_pre^2) + (N_Exp_pre-1) * SD_Exp_pre^2)) / (N_Ctrl_pre + N_Exp_pre – 2))

ES_BG = Mean change_BG / SD* pooled_BG

Legend. ES = Effect size, WG = Within group, BG = Between group, Exp = experimental group, Ctrl = control group, N = sample size, pre = pre-intervention measure, post = post-intervention measure, SQRT = square root, ^2 = raised to the power of two.

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Supplement 3. Effect sizes for outcome measures and sub-measures

Name, n pre/post	Measure	Within group absolute mean change (post - pre experiment)	SD* Exp pooled	ES Hedges' g Exp (Delta mean/ SD* pool)	Between group absolute mean change (experiment – control group)	SD* Baseline pooled	ES Hedges' g (Delta Mean / SD* pool)
With control group							
AMUTIO* n=21/21	FFMQ	0.37	0.48	0.78	0.54	0.36	1.51
	Observe	0.65	0.62	1.05	0.84	0.75	1.12
	Describe	0.25	0.67	0.37	0.31	0.63	0.49
	Awareness	0.32	0.77	0.42	0.34	0.79	0.43
	Non-judgement	0.36	0.65	0.55	0.39	0.58	0.67
	Non-reactive	0.29	0.57	0.51	0.26	0.54	0.48
	SRSI-3 Relaxation	0.54	0.57	0.95	0.46	0.52	0.89
	SRSI-3 Pos energy	0.71	0.74	0.97	0.71	0.63	1.13
	SRSI-3 Mindfulness	0.71	0.81	0.88	0.76	0.91	0.84
	SRSI-3 Transcend	0.6	0.93	0.65	0.84	0.88	0.96
VERWEIJ** n=43/43 exp, n=20/20 control	FFMQ	6.87	14.49	0.47	6.9	13.82	0.50
	Observe	2.04	4.49	0.45	1.56	4.28	0.36
	Describe	1.51	4.69	0.32	1.64	4.52	0.36
	Awareness	0.98	4.59	0.21	1.57	4.47	0.35
	Non-judgement	1.2	5.25	0.23	1.37	5.04	0.27
	Non-reactive	1.13	3.71	0.30	0.76	3.56	0.21
	JSPE total	1.95	9.90	0.20	0.2	9.61	0.02
ASSUERO n=43/43 exp n=25/25 control	FFMQ	12.40	19.50	0.64	11.3	19.65	0.57
	Observe	4.30	7.07	0.61	4.7	7.33	0.64
	Describe	0.70	5.50	0.13	0.6	5.52	0.11
	Awareness	2.10	5.33	0.39	2.7	6.24	0.43
	Non-judgement	3.50	7.05	0.50	2.4	7.35	0.33
	Non-reactive	2.20	3.71	0.59	1.8	3.85	0.47
	JSPE total	3.50	11.32	0.31	5.3	12.10	0.44
	Perspective	1.30	5.88	0.22	1.5	6.62	0.23
	Compassion	1.80	4.54	0.40	2.3	5.03	0.46
	In patients shoes	0.40	1.49	0.27	1.5	1.54	0.98
DYRBYE n=40/40 exp n= 39/39 control	CD-RISC-10	1.30	5.44	0.24	0.7	5.46	0.13
	PJSS	1.00	9.94	0.10	0.6	10.11	0.06
	UWES vigor	0.30	0.94	0.32	0.2	1.04	0.19
	UWES dedication	0.10	0.87	0.11	-0.1	0.96	-0.10
	UWES absorption	-0.10	0.88	-0.11	-0.2	1.00	-0.20
	Empowerment at work	2.7	11.29	0.24	-0.3	12.56	-0.02
GARDINER n=86/77 exp, n=24/19 control	WRM-7 total	2.79	6.46	0.43	2	6.63	0.30
	QoWL-6 total	2.92	6.50	0.45	1.76	6.48	0.27
SCHROEDER N=16/13 exp, n=17/13 control	MAAS	0.62	0.99	0.63	0.76	0.86	0.88
	BRS	2.53	4.93	0.51	2.95	4.81	0.61
	SCBC	1.53	4.33	0.35	3.46	4.75	0.73
MARGALIT*** n=44/44	Self-esteem	4.4	13.72	0.32			

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Without control group					
MONTERO-MARIN n=30/30	PANAS 1 session/wk	1.25	6.10	0.20	
	PANAS 2 session/wk	2.97	5.69	0.52	
	MAAS 1 session/wk	2.67	11.49	0.23	
	MAAS 2 session/wk	4.59	12.28	0.37	
	CD-RISC total 1 session/wk	1.23	7.31	0.17	
	CD-RISC total 2 session/wk	2.48	6.79	0.37	
VON WIETMARSCHEN n=50/50	SCS	0.5	0.65	0.77	-
	GRAS	3.3	7.22	0.46	-
FORTNEY n=30/28	RS-14	2.1	12.28	0.17	-
	SCBC	-0.2	4.54	-0.04	-
KRASNER n=60/59	JSPE total	4	9.16	0.44	-
	Perspective	2	5.79	0.35	-
	Compassion	1.6	4.24	0.38	-
	In patients' shoes	0.8	2.12	0.38	-
	Mindfulness 2 factor				
	FFMQ (BAER) total	7.7	7.33	1.05	-
	BAER observe	5	4.63	1.08	-
	BAER non-react	3.2	3.95	0.81	-
REES n=7/7	WHO-5 (WB)	10.3	19.79	0.52	-
	PANAS-pos	-			-
WHALLEY n=2081/1345	JSS WCW	0.59	1.35	0.44	-

Notes: Amutio*, no information given on pre-post sample size, assumption made that there was no attrition. Verweij**, wait list control participants were included in both intervention and control groups. *** Margalit only pooled group (experimental and control) analysis provided. For the FFMQ Verweij and Assuero have much larger numbers than Amutio, this is presumably due to reporting of the total score count as opposed to the total mean scores.

Outcome measures: JSS WCW Warr Cook Wall Job satisfaction Scale; JSPE Jefferson Scale of Physician Empathy; FFMQ Five Facet Mindfulness Questionnaire; Baer (2 item version of FFMQ); SRSI Smith Relaxation States Inventory; SCS Self Compassion Scale (Neff); FR-JS Font Roja Job Satisfaction Questionnaire; SCBC Santa Clara brief compassion scale (an abbreviation of the Sprecher and Fehr's Compassionate Love Scale); PJSS Physician Job Satisfaction Scale (3 dimensions JS, career satisfaction, and specialty satisfaction), 12-item Global Job Satisfaction subscale used; CD-RISC Connor-Davidson Resilience Scale; PANAS Positive And Negative Affect Scale, MAAS Mindful attention awareness scale; BRS Brief Resilience Scale; GRAS Groningen Reflection Ability Scale; RS-14 Resilience, WHO-5 Wellbeing Index, WRM Work Related Morale Measure. QoWL-6 Quality of Work Life.

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Supplement 4. Outcome measures utilized in the included studies.

Acronym	Outcome measure	No of studies used in
ACP/ASIM	Physician satisfaction (Am College of Phys / Am Soc of Int Med)	1
BAER	Baer mindfulness scale (2 item version of FFMQ)	1
BRS	Brief resilience scale	1
CD-RSIC-10	Connor-Davidson Resilience Scale	2
FFMQ	Five Facet Mindfulness Questionnaire	3
FR-JS	Font Roja Job Satisfaction scale	1
GRAS	Groningen Reflection Ability Scale	1
JSPE	Jefferson Scale of Physician Empathy	3
JSS WCW	Job Satisfaction Scale Warr-Cook-Wall	2
LS	Life satisfaction	1
MAAS-15	Mindful attention awareness scale	2
PANAS	Positive and negative affect scale	2
PJSS	Physician Job satisfaction scale	1
PWS (MEMO)	Physician Workplace Satisfaction	2
QoWL-6	Quality of work life	1
RS-14	Resilience	1
SCBC	Santa Clara brief compassion scale	2
SCS	Self compassion scale (Neff)	1
SRSI-3	Smith Relaxation states	1
UWES	Utrecht work engagement score	1
WHO-5	Wellbeing Index	1
WRM-7	Work related morale measure	1
	Self-esteem	1

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The PRISMA for Abstracts Checklist

TITLE	CHECKLIST ITEM	REPORTED ON PAGE #
1. Title:	Identify the report as a systematic review, meta-analysis, or both.	1
BACKGROUND		
2. Objectives:	The research question including components such as participants, interventions, comparators, and outcomes.	2
METHODS		
3. Eligibility criteria:	Study and report characteristics used as criteria for inclusion.	2-4
4. Information sources:	Key databases searched and search dates.	
5. Risk of bias:	Methods of assessing risk of bias.	
RESULTS		
6. Included studies:	Number and type of included studies and participants and relevant characteristics of studies.	4-8
7. Synthesis of results:	Results for main outcomes (benefits and harms), preferably indicating the number of studies and participants for each. If meta-analysis was done, include summary measures and confidence intervals.	
8. Description of the effect:	Direction of the effect (i.e. which group is favoured) and size of the effect in terms meaningful to clinicians and patients.	
DISCUSSION		
9. Strengths and Limitations of evidence:	Brief summary of strengths and limitations of evidence (e.g. inconsistency, imprecision, indirectness, or risk of bias, other supporting or conflicting evidence)	8-12
10. Interpretation:	General interpretation of the results and important implications	
OTHER		
11. Funding:	Primary source of funding for the review.	13
12. Registration:	Registration number and registry name.	

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The effect of interventions for the wellbeing, satisfaction and flourishing of general practitioners – A systematic review

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TITLE PAGE

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ABSTRACT

Objectives
Clinician wellbeing has been recognised as an important pillar of health care. However, research mainly addresses mitigating the negative aspects of stress or burnout, rather than enabling positive aspects. With the added strain of a pandemic, identifying how best to maintain and support the wellbeing, satisfaction and flourishing of general practitioners (GPs) is now more important than ever.

Design
Systematic review.

Data sources
We searched MEDLINE, PsycINFO, Embase, CINAHL, and Scopus from 2000 to 2020.

Study selection
Intervention studies with more than 50% GPs in the sample evaluating self-reported wellbeing, satisfaction and related positive outcomes were included. The Cochrane Risk of Bias 2 tool was applied.

Results
We retrieved 14,792 records, 94 studies underwent full text review. We included 19 studies in total. Six randomised controlled trials, three non-randomised, controlled trials, eight non-controlled studies of individual or organisational interventions with a total of 1141 participants. There were two quasi-experimental articles evaluating health system policy change.

Quantitative and qualitative positive outcomes were extracted and analysed. Individual mindfulness interventions were the most common (k=9) with medium to large within (0.37-1.05) and between group (0.5-1.5) effect sizes for mindfulness outcomes, and small to medium effect sizes for other positive outcomes including resilience, compassion, and empathy. Studies assessing other intervention foci or other positive outcomes (including wellbeing, satisfaction) were of limited size and quality.

Conclusions
There is remarkably little evidence on how to improve GPs wellbeing beyond using mindfulness interventions, particularly for interventions addressing organisational or system factors. This was further undermined by inconsistent reporting, and overall high risk of bias. We need to conduct research in this space with the same rigour with which we approach clinical intervention studies in patients.

Registration
The systematic review protocol is registered on PROSPERO CRD42020164699.
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ARTICLE SUMMARY

Strengths and limitations of this study

- Whilst burnout has been a general focus of research, interventions to improve the wellbeing of general practitioners (primary care physicians) appear sparse.
- The strength of this study is the extensive and systematic approach taken to evaluate interventions aimed at improving wellbeing, satisfaction, flourishing, and other positive outcomes in general practitioners.
- The systematic review was registered on PROSPERO a priori before commencing the data selection and extraction process.
- English only articles were included.
- The limitation of this systematic review is the dependency on the number of retrieved and included publications, and their quality of methodology and reporting.

INTRODUCTION

Mental ill-health, burnout and stress amongst health care practitioners are a huge concern internationally with impacts on individual doctors and their families, patient care, and the sustainability of the health care system (1). The wellbeing of clinicians has been recognised as crucial, and has been added to the more commonly shared health system goals of: improved care for individuals, better population health, and reduced health care costs (2-5). Despite this stated aim, few studies have evaluated interventions to improve wellbeing, satisfaction, and flourishing in general practitioners (GPs) (6-12), typically taking a more traditional, problem-focused approach, such as investigating causes and reduction of burnout and stress.

Medical doctors that provide primary care to patients are the backbone of health care provision. In Australia, the UK and Europe, typically the term ‘general practitioner’ (GP) is used, whereas North American articles generally refer to ‘primary care physicians’ or ‘family doctors’. Terms are used interchangeably.

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Satisfaction and wellbeing in GPs are known to be associated with several factors. In the USA, primary care physicians are less satisfied with their job than specialists (13, 14). A ten-year decline in job satisfaction for British GPs has been reported (15), and a Norwegian longitudinal study reported dwindling GP satisfaction over seven years (16). In primary care, job satisfaction correlates with practice resources, an ordered atmosphere, a practice culture that enables communication, and ease of providing quality care (13, 17, 18), and is inversely related to isolation and low sense of community (19, 20).

For wellbeing, Murray et al. conducted a cross-sectional study exclusively exploring positive mental health and psychological resources (wellbeing, resilience, self-efficacy, and hope) of GPs (21). GPs positive mental health was comparable to the general population and older and female GPs fared best, suggesting interventions for younger and male GPs may be most useful.

Systematic reviews and interventions aimed at improving satisfaction and wellbeing in GPs appear sparse. A European collaboration conducted a systematic review and qualitative study looking at positive determinants of satisfaction in GPs. They identified general work-related factors (i.e., workload, responsibility, recognition, and income), and GP specific factors (i.e., competence, intellectual stimulus, and work-life balance) (22, 23). However, there does not appear to be a systematic review looking at interventions to improve satisfaction exclusively in GPs.

A systematic review of interventions to improve the psychological wellbeing of GPs identified only four articles; two cognitive-behavioural, one mindfulness-based intervention and one self-help information approach (24). They applied a dual model of languishing/flourishing and the presence of mental illness/absence of mental illness matrix (25-28).

Overall, little seems known about which interventions are efficacious in promoting GPs wellbeing and satisfaction. In contrast to more extensive research on burnout, distress, and mental ill health with a view to treat, avoid or mitigate negative outcomes, we explicitly aimed to apply a positive lens and focus on interventions that enhance GPs' satisfaction and wellbeing, or promote environments and individual behaviours that may

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better enable wellbeing. We purposely included interventions on any level; directed at the individual (i.e., training, workshops), the organisation (i.e., work-flow improvements in the practice) and on a systemic level (i.e., policy change). It is useful to bear in mind that GPs are typically high functioning individuals and are not a priori a clinical population, which is why we believe a positive framework is most preproperate. Currently, reports indicate that COVID-19 places enormous additional strain on health professionals which impacts their physical, mental, and social wellbeing (29, 30). GPs as the first port of call may be particularly affected, whilst playing a crucial role in supporting population health (31-33). Efforts must be made to understand how GPs remain well, and if necessary, put measures in place to assist this.

OBJECTIVES

We systematically reviewed studies exploring the effect of any type of intervention on the wellbeing, satisfaction, and wellbeing of GPs. We broaden and expand upon the existing literature, deliberately including any type of intervention, and a range of positive outcomes, and explore if there have been more recent intervention studies conducted in this field.

METHODS

Data sources and search strategy

We followed PRISMA guidelines (34), and consulted a specialist librarian (JKH). MEDLINE, PsycINFO, Embase, CINAHL and Scopus were searched on 13th January 2020.

PICO search terms included GPs and synonyms; interventions and implementations directed at the individual, the level of the organisation or practice and the health care system; outcomes included wellbeing, satisfaction, flourishing, synonyms and antonyms (*search strategy, supplement 1)*. Titles, abstracts, text, key terms, and subject headings were searched for English publications. Eligible articles and related systematic reviews were hand-searched for further relevant references, and authors were asked to supply full text articles where relevant conference abstracts only were available.

Study selection and data extraction

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Due to the large number of records, sample screening of 1% of titles (n=107) was performed and discussed by two reviewers (DNA, AS) together, and excellent agreement was reached. DNA screened all titles. DNA and AS independently and separately examined all abstracts and full-text records for inclusion using Covidence (35). Initial inter-rater reliability showed a proportionate agreement of 0.94 with Cohen's Kappa of 0.68 for abstracts. For full text screening there was a proportionate agreement of 0.84 with Cohen's Kappa 0.56, whereby both reviewers agreed to include 15/94 full text articles and exclude 64/94 articles. Initial disagreements on 15 articles were resolved through joint discussion, or third reviewer adjudication (NG).

Studies with more than 50% GPs (family doctors, primary care physicians) working in a practice setting or medical centre, reporting on wellbeing, satisfaction, flourishing, mindfulness, resilience, empathy, engagement, balance, empowerment, compassion, work-related morale, and control measures were included. We excluded studies exclusively reporting on burnout, distress, or mental ill-health.

Data including author, year, type of study, participants, intervention, pre- and post-outcome measures, and quantitative and qualitative results were extracted. *(Table 1).*

Data synthesis and analysis

We calculated within- and between-group absolute change and effect sizes (Hedges' g) *(see table 2, 3 and supplement 2, 3)* (36). We compared mean outcome scores and standard deviations (SDs) at baseline with post-intervention scores. Where several post-intervention measures were reported, the primary outcome point nominated by the authors was selected. We utilised SD*, which takes different sample sizes into account *(formulae in supplement 2).* For within-group we calculated the pooled SD* based on pre- and post- interventions SDs, for between-group analysis the effect size was calculated based on the pooled SD* of control and intervention group at baseline (37) *(supplement 2).*

Positive effect sizes indicated an effect for the intervention. Effect sizes of 0.2, 0.5 and 0.8 were considered small, moderate and large, respectively (38).

Risk of bias

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Two reviewers (AS, DNA) independently applied the Cochrane RoB2 (39) to RCTs. Total RoB2 scores showed 100% agreement. Any discrepancies of sub-scores were discussed, and consensus was achieved. The other studies were assessed by DNA.

Patient and public involvement

No patients are involved.

RESULTS

The database searches rendered 14,792 records in total. After removing duplicates, studies conducted before the year 2000, and adding 12 studies through hand search - which included contacting authors for full text papers of relevant conference abstracts - 10,759 studies were screened. We eliminated 9,682 records by title, and 983 by abstract, leaving 94 studies for full-text assessment.

Study characteristics & design

We included nineteen studies in the systematic review (2, 10, 40-56) *(Table 1 and 2, and PRISMA-Flowchart Figure 1)*. Six randomised controlled trials (RCT), three non-randomised controlled trials (CBA), eight non-controlled interventions (NCBA), and two reports from a longitudinal cohort during which a health policy change was introduced, which we considered as 'naturalistic' interventions (41, 56). Five studies included a qualitative component. Only one RCT (50) and two CBAs (44, 47) utilised active controls. Five RCTs and one CBA (40) had a waitlist control arm. Publications from the United States (8/19, 42%), Europe (8/19, 42%), Australia (2/19) and Israel (1/19) were included *(Table 1)*.

Participants

The total population enrolled was 1,141 for the 17 intervention studies (average participants per study 67.1, range 6 – 290). The two studies reporting on the same panel survey population (41, 56) included approximately 2,000 participants each year. Mean age overall ranged from 40 – 54.9 years, and sex from 8% - 76.9% male participants *(Table 1)*. Attrition for intervention groups varied from 0% to 20%, for controls from 0% to 24%. One outlier had a total attrition rate of 80% (51). Eight studies reported

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follow-up measures, timepoints ranged from 3 to 48 months post baseline (mean 14.6 months).

Intervention type

We found considerable variation in intervention type, length, and dose-intensity. Three groups were distinguished based upon the focus of the intervention: individual/personal (13/19, 68% of studies, N=930), organisational (4/19, 21%, N=211) and naturalistic interventions on a systemic level (policy change in the UK) (2/19, 11%).
Individual mindfulness-based interventions were most common (9/13, 69%), followed by educational training or experiential workshops (3/13, 23%) with one coaching intervention (46). Two organisational interventions trialled the addition of clerical support or scribes, and two explored an organisational improvement programme. Two studies from the United Kingdom examined the effects of the introduction of a pay for performance scheme (41, 56) *(Table 2)*.

Outcomes and their measures

The definitions of outcomes and measurement tools varied considerably. Only one study clearly stated one a priori primary outcome (51), with most using a battery of self-reported outcome measures *(supplement 4)*. These included a range of twelve validated tools (BAER, BRS, CD-RISC-10, FFMQ, GRAS, JSPE, JSS-WCW, MAAS, PANAS, SCS, UWES, WHO-5) (57-75) as well as thirteen measures where no validation information was obtainable. A range of job satisfaction measures were applied in eight studies, mindfulness in six, resilience in four, compassion and empathy tools were each used thrice, the positive and negative affect scale was used twice. The WHO-5 wellbeing index was used once *(supplement 4)*. Not one study evaluated flourishing. Negative outcome measures were often concurrently reported. Sixteen studies employed the Maslach Burnout Inventory or other stress-related measures. As the a priori aim of the study was to explore effects of interventions on wellbeing, satisfaction, and other positive outcomes, we did not extract and report results for negative outcome measures, nor examine possible inter-relationships between positive and negative outcome measures.

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Intervention effects

The between-group change for controlled studies and within-group change for intervention arms are presented in *Tables 2, 3 and supplement 3.*

a. *Individual focussed interventions*

i) Mindfulness (k=9)

Six mindfulness interventions (3 RCTs, 1 CBA, 2 NCBA) evaluated mindfulness outcomes (FFMQ, MAAS, BAER) and reported moderate to high between-group effect sizes (k=4) ranging from 0.5 – 0.88 for mindfulness with an outlier at 1.5 (42). Within-group ES (k=6) showed moderate effect sizes (range 0.47 – 0.78) with one study outlier at 0.37 and one at 1.05 (*Tables 2 and 3*).

Studies frequently utilised resilience, compassion, and empathy measures with overall low to moderate effect sizes. One RCT and two NCBAs measured resilience (BRS, RS-14, CD-RISC), whereby between-group ES (k=1) was moderate at 0.61, while within-group (k=3) effect sizes were low to moderate (range 0.17 – 0.51). Compassion measures (SCBC, SCS) were reported in three studies (1 RCT, 2 NCBAs). Between-group ES (k=1) was 0.73, whilst within-group ES (k=3) varied considerably (-0.04 to 0.77). Three studies reported on empathy (JSPE) (1 RCT, 1 CBA, 1 NCBA) with very low 0.02, respectively moderate between-group 0.44 ES (k=2), and within-group ES ranging from 0.2 - 0.44 (k=3) *(Tables 2 and 3).*

Two mindfulness studies (NCBAs) measured positive affect (PANAS), only one reported a within-group ES (0.52). One NCBA reported a within-group ES (0.52) for wellbeing (WHO-5), another NCBA reported an ES of 0.46 for self-reflection.

These effect sizes are generally supported by the results reported in the individual studies *(Table 1).* Several interventions included repeated measures at later time-points, i.e., during maintenance phase (48), showing an ongoing impact of mindfulness practice. Qualitative results suggested increased wellbeing and compassion towards self and others (40), respectively, improved awareness, acceptance, peacefulness, and openness (55) after the intervention.

ii. Training, workshops, and coaching (k=4)

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For training, workshops and coaching interventions, we were only able to obtain data to calculate the ES of one RCT (46) and one CBA (47). Low between-group effect sizes for work-related morale (0.3), quality of work-life (0.27), and low ES for both measures within-group (0.43 and 0.45 respectively) were found for Gardiner's CBA (47). Very low effect sizes for job satisfaction and resilience both for between-group (0.06, 0.13), and within-group (0.13, 0.24) change were observed in Dyrbye's RCT (46).
These results are reflected in the individual study results *(Table 1)*. Barcons (44) did not detect any significant between-group differences for overall job satisfaction, whilst Margalit (50) demonstrated significant improvement in self-esteem between groups.

b. Organisational interventions (k=4)

One RCT and three NCBAs trialled organisational interventions. Means and standard deviations were not provided; therefore, we were not able to calculate effect sizes. Linzer et al. (RCT, n=166) demonstrated that workflow interventions, communication and overall quality improvements benefited satisfaction in the intervention group (49). Whilst Dunn et al. (NCBA, n=32), showed that quality improvement projects in the workplace showed significant improvement in quality work competence ratings but fluctuating satisfaction levels (2). Two smaller uncontrolled trials (n=13 in total) investigated the addition of clerical staff to the practice. Pozdnyakova et al. showed that the addition of clerical staff led to an improvement in satisfaction with the clinic workflow from 2/6 to all 6 GPs in a single practice but did not report on any other measures of wellbeing (52). Similarly, Contratto et al. reported improved quality of life and professional balance for seven general medicine physicians in a mixed-methods approach (45).

c. Systemic interventions (k=2)

The introduction of a new contract with pay for performance scheme showed a significant improvement in job satisfaction (56) with an effect size of 0.44 between 2004 and 2005. Allen et al. used the same data and included a 2008 survey to look at satisfaction as a function of the exposure of GPs to the pay per performance scheme. Whilst job satisfaction declined again in 2008, the exposure to the scheme didn't affect satisfaction (41).

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Risk of bias

The types of intervention and study settings precluded blinding for randomised controlled studies (no allocation concealment for waitlist control groups), and the outcome measures were participant reported throughout, and as such all studies were rated as high risk of bias by the Cochrane RoB2.

DISCUSSION

Strengths, limitations, and interpretation of evidence

We identified nineteen studies, half of which were published in the last four years, demonstrating an increased interest in the improvement of wellbeing and satisfaction of GPs. In comparison, a systematic review from 2016 looking at interventions to reduce burnout in physicians included fifteen RCTs and 37 cohort studies with 20 studies conducted before 2010 (76), suggesting burnout has been a research focus for longer. Or this may possibly indicate the focus is more generally shifting from a disease and 'dis-abled' to an 'en-abled' approach when trying to design interventions for health care professionals.

The considerable heterogeneity in the definition and measurement of constructs, study design, participant numbers, intervention types, intervention dose (ranging from six to 53 hours), follow-up periods, quality and reporting precluded a meta-analysis. It is challenging to draw robust conclusions regarding the (comparative) effectiveness of the different types of interventions reviewed.

Mindfulness interventions provided the most comprehensive and robust data with moderate to large effect sizes on mindfulness outcomes, and low to moderate effect sizes on compassion, resilience, and empathy. Our results are supported by two reviews looking at the effects of mindfulness-based interventions on wellbeing (77, 78), in health care professionals more generally. Lomas et al. conducted a systematic review and meta-analysis and assessed 41 studies with approximately 2,100 participants. They found a wide range of self-reported outcomes (with both positive and negative measures of wellbeing). Reported within-group effect sizes for mindfulness, positive wellbeing (or life satisfaction), and compassion hovered around a moderate 0.5 mark, ES for

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empathy was 0.31; whilst for RCTs, the between-group ES for mindfulness, life satisfaction, and compassion were around 0.3 (77).
Scheepers et al. contributed a narrative review of 23 studies looking at mindfulness-based interventions for wellbeing in doctors of all ages and specialities. Review authors noted some caveats; considerable variation in type and dose-intensity of practice, sparse long-term outcome data, and methodological limitations. They cautioned that mindfulness practice involves time and dedication, which is not always feasible for busy health care professionals. In sum, the conclusions they drew are tentatively positive.
In contrast to Lomas et al. our ES for mindfulness is higher between-group than within-group, which is somewhat unusual. This may be explained by one study (42) whose positive outcome appeared determined by the decline in mindfulness over time in their control group, rather than the intervention being effective.

We identified four studies evaluating coaching and experiential workshops for GPs, which showed low effect sizes for satisfaction measures and moderate ES for work-related morale and quality of work life. There does not appear to be much literature on coaching for health care professionals. One quasi-experimental study by Gardiner et al. looked at 'cognitive behavioural coaching' in rural Australian GPs and demonstrated a significant within-group reduction in distress levels for the coachees (79). Resilience training for a range of different physicians that had completed training was investigated in a recent systematic review. Four RCTs and five observational studies were included. The authors flag heterogenous study design and use of outcome measures, as well as quality issues with weak evidence for the interventions, whilst indicating potential for improvement of resilience (80).

We found four small-scale organisational interventions that suggested improved (job) satisfaction, as did one large-scale health policy intervention of performance-related pay in the UK. For burnout, a paucity of interventions trials delivered at organisational and systemic levels has been previously reported (81, 82), the authors suggest to actively design such trials. Similarly, Dyrbye et al., concluded that whilst useful, an individual intervention such as coaching is no replacement for organisational improvement (46). Shanafelt and colleagues have collated their vast research into burnout and put forward

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nine organisational strategies to address burnout and physician wellbeing through leadership (83). Despite calls for action, these avenues have not been adequately addressed or reported to date, at least not for GPs, and warrant further exploration. Considering the time it takes to gather and report data, it is understandable that organizations might feel pressure to implement programs based on preliminary data. Commendably, Dutch researchers recently investigated the effects of a mindful leadership course in hospital-based medical specialists (84, 85). Both a qualitative interview and a pre- and post- self-evaluation suggested an overall benefit of the intervention with improved mindfulness, life satisfaction and leadership, reduced burnout, and positive change in attitudes and behaviours towards self and others. Not all participants benefited equally, suggesting a need to provide a range of interventions to meet defend participants' needs. Future investigation will need to explore what the impact on individuals' leadership style and on their teams is.

Limitations

We included English publications only, although purposely extending our search globally. We excluded studies before the year 2000, because wellbeing literature in medicine is a more recent development, and general practice is now likely quite different than two decades ago. Self-reported outcome measures are typically subject to bias, particularly considering studies included GPs from different settings and cultures, potentially introducing cultural bias, rendering comparisons challenging.

Suggestions for future research

Based on our findings, we provide some suggestions which may be useful for future research into wellbeing and satisfaction for GPs. Stronger collaboration amongst researchers in this space may also lead to improved results.

a) Selection of outcomes & outcome measures

The reduction in burnout and stress is often equated with an improvement in wellbeing or satisfaction. We argue that the improvement of negative outcomes doesn't necessarily indicate a presence of satisfaction or wellbeing. This aligns with the dual continuum model of mental health / mental ill-health and flourishing / languishing (25-

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28). Good mental health isn't automatically linked to flourishing, nor is mental ill-health an indicator of languishing. Other authors have made similar statements (10, 24, 42, 77). We did not find a single study about flourishing in GPs.

Clearly defining the constructs 'wellbeing' and 'satisfaction', whilst utilising validated wellbeing and satisfaction measures, will enhance clarity, consistency and comparability of study design and reporting. We suggest drawing on existing frameworks, models, and definitions in the psychological literature (for different types of wellbeing, satisfaction or flourishing) (86-89). To measure wellbeing we suggest the Warwick Edinburgh Mental Wellbeing Scale (WEMWBS) and for Job satisfaction the Warr-Cook-Wall scale (WCW-JS), both of which have been validated in medical populations (90, 91). Brady et al., who conducted a systematic review looking at the definition and measurement of 'physician wellness' similarly stated that there needs to be consensus and clarity of definition, if we want to improve the quality and comparability of research in this space (92). Whilst this would improve the next phase of studies, the urgency in calls for actions may need to be balanced against the calls for consistency among studies.

b) Organisational and systemic interventions

With the dearth of research in this space, and the relatively small effects for individual interventions, we believe it worthwhile to explore system and organisational interventions (i.e. mindful leadership training describe above) in the context of wellbeing.

Considering what is known about burnout (drivers being organisational culture, workplace conditions, lack of control and autonomy), it's not surprising that individual interventions are not as effective as desired (49, 81, 93-95). Hence, more combined approaches targeting both individuals and organisations have been proposed.

A 2017 British meta-analysis contrasted different types of interventions for burnout on the individual doctor and on the systemic level, whereby systemic interventions appear more effective (96). Similarly, groups in the USA state that the approach must be combined and include organisational interventions (1, 97, 98), mostly focusing on time management, rostering, workflow management, staffing and use of information technology solutions. Overall, there is a scarcity of organisational interventions aimed at

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reducing burnout (99), and conclusions from the two meta-analyses of interventions to reduce burnout should be considered provisional.

In sum, we endorse an intensified effort to explore organisational interventions to improve wellbeing and satisfaction, and believe a focus on leadership and improving the culture at work is a good place to start.

c) Physical interventions

We didn't find any physical interventions (i.e. exercise, nutrition) geared towards improving GPs' wellbeing. Sparse research on exercise or diet interventions for doctors exists. A Pakistani cross-sectional survey revealed that 76% of nearly 1200 doctors, nurses and dentists did not exercise at all, and only one participant ate according to USDA dietary guidelines (100). While a US cross-sectional survey of 303 physicians found that less than 25% knew the American Heart Association (AHA) dietary recommendations, whilst around half knew and followed their physical activity guidelines (101). Two systematic reviews looked at exercise and burnout in the general population, one was inconclusive (102), the other stated that physical activity effectively reduces burnout (103). Both identified methodological issues and no long-term follow-up. Seeing the paucity of data, this might provide an avenue for further investigation.

d) Quality and risk of bias

Areas for risk of bias are inherent in this type of research. However, measures can be taken to reduce bias for example by using active controls in randomized studies as suggested by other review authors (78), by consistently publishing study protocols a priori, and controlling for participant attributes, such as prior engagement in mindfulness practice. Ideally, the same rigorous approach should be applied to intervention studies for clinicians, as to clinical interventions studies for patients.

CONCLUSION

Despite increasing interest in the improvement of wellbeing and satisfaction, there is remarkably little evidence beyond mindfulness interventions aimed at individual GPs. Few studies utilize validated measures of wellbeing and satisfaction, and favour burnout

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tools. Studies looking into organisational and systemic interventions remain sparse, and conclusions about their effectiveness may be premature.

Considering the COVID-19 pandemic and the added strain to primary care, programmes to support and research GP wellbeing should be prioritised by policymakers and governments worldwide.

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Registration

The protocol is registered on PROSPERO CRD42020164699.

Ethics approval statement

We did not seek ethics approval, as this is a systematic review of previously published data, that is available in the public domain.

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Competing interest statement

All authors have completed the Unified Competing Interest Form.

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Ian Hickie has declared financial relationships outside of the submitted work in the previous three years and other relationships or activities.

Authors' Contributions

DNA is the guarantor and corresponding author and attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted.

For peer review only

Authors contributed to the study conception and design (DNA, JKH, RE, IH, NG), the acquisition (DNA, AS, NG), analysis (DNA, AS, NG) and interpretation (DNA, AS, RE, IH, NG) and the drafting or critical revision of important intellectual content and final approval of the version to be published and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Transparency declaration

Dr Diana Naehrig (the manuscript's guarantor) affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned and registered have been explained.

Data sharing statement

All included studies are published. We will consider sharing data upon reasonable request.

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Figure 1. Prisma diagram

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Table 1. Evidence table

Author, Date, Study design	Partici-pants (n, mean age, % male)	Interven-tion (n, mean age)	Control (n, mean age)	Positive outcome measure	Other outcome measure	Baseline scores	Post-intervention score	Follow-up score (mean, SD, n)	Results - Summary
Allen 2017 Panel survey linked with QOF data 2004/5 (first year) and 2007/8 (fourth year)	GPs in the UK. 2004: n=1950, 47.0yrs, 66.2% male. 2005: n=2000, 47.9yrs, 63.6 % male. 2008: n=1986, 48.7yrs, 63.3% male.	Pay for perform-ance (P4P) scheme; Quality and Outcomes Framework (QOF) introduced in NHS in 2004	NA	Job satisfacti on (JSS WCW). Life satisfacti on	P4P exposure	Mean. 2004: Life satisfaction (4.649). Overall JS (4.567), Physical working conditions (4.862), Choose method of working (4.636), Colleagues (5.515), Recognition for good work (4.224), Responsibility (4.976), Remuneration (4.376), Opportunity to use abilities (4.787), Hours of work (3.914), Variety in job (5.011).	Time at 12 months. Mean. 2005: Life satisfaction (5.095). Overall JS (5.201), Physical working conditions (5.044), Choose method of working (4.892), Colleagues (5.599), Recognition for good work (4.726), Responsibility (5.406), Remuneration (5.387), Opportunity to use abilities (5.147), Hours of work (4.802), Variety in job (5.269).	Time at 48 months. Mean. 2008: Life satisfaction (5.008). Overall JS (4.728), Physical working conditions (5.129), Choose method of working (4.640), Colleagues (5.602), Recognition for good work (4.495), Responsibility (5.276), Remuneration (4.849), Opportunity to use abilities (5.074), Hours of work (4.205), Variety in job (5.276).	Between 2004 and 2005 overall job satisfaction increased (also see Whalley 2008), then fell from 2005 to 2008. There is a positive effect (though not statistically significant) between QOF/P4P income exposure and job satisfaction in 2005 (t-ratio 1.74), though not so in 2008 (t-ratio 0.14). The P4P exposure shows now significant effect on GPs job satisfaction.
Amutio 2015 RCT	Physic-ians (approx 70% primary care) in public or private practice in Spain (n=42, 47.3 yrs, 42.9%)	8-week MBSR programme (group course 8x2.5h/wk, 1x8h retreat plus homework), 10-month mainten-ance period (group course 10x2.5h/mt plus homework) (n=21)	WLC (n=21)	Mindful-ness (FFMQ), Relax-ation (SRSI-3)	Heart rate	Mean and SD. INTERVENTION (n=21) FFMQ Total (3.34, 0.44), FFMQ Observing (3.33, 0.60), FFMQ Describing (3.58, 0.72), FFMQ act with awareness (3.16, 0.87), FFMQ Nonjudging (3.42, 0.64), FFMQ Nonreactivity (3.17, 0.51). SRSI relaxation (2.54, 0.53), SRSI positive energy (3.09, 0.64), SRSI mindfulness (3.74, 0.89), SRSI Transcendence (2.67, 0.82). CONTROL (n=21) FFMQ Total (3.51, 0.25), FFMQ Observing (3.02, 0.88), FFMQ Describing (3.88, 0.53), FFMQ Act with awareness (3.93, 0.70), FFMQ Nonjudging (4.19, 0.52), FFMQ Nonreactivity (3.23, 0.57). SRSI relaxation (2.52, 0.5), SRSI positive energy (3.01, 0.62), SRSI mindfulness	Time at 8 weeks. Mean and SD. INTERVENTION (n=21) FFMQ Total (3.71, 0.51), FFMQ Observing (3.98, 0.64), FFMQ Describing (3.83, 0.62), FFMQ act with awareness (3.48, 0.65), FFMQ Nonjudging (3.78, 0.66), FFMQ Nonreactivity (3.46, 0.63). SRSI relaxation (3.08, 0.61), SRSI positive energy (3.80, 0.82), SRSI mindfulness (4.45, 0.71), SRSI Transcendence (3.27, 1.02). CONTROL (n=21) FFMQ Total (3.34, 0.33), FFMQ Observing (2.83, 0.98), FFMQ Describing (3.82, 0.58), FFMQ Act with awareness (3.91, 0.61), FFMQ Nonjudging (4.16, 0.52), FFMQ Nonreactivity (3.26, 0.67). SRSI relaxation (2.60, 0.51), SRSI positive energy (3.01, 0.6),	Time at 12 months. Mean and SD. INTERVENTION FFMQ Total (3.85, 0.49), FFMQ Observing (4.09, 0.62), FFMQ Describing (4.01, 0.58), FFMQ act with awareness (3.55, 0.69), FFMQ Nonjudging (3.96, 0.66), FFMQ Nonreactivity (3.58, 0.55). SRSI relaxation (3.16, 0.78), SRSI positive energy (3.99, 0.81), SRSI mindfulness (4.60, 1.01), SRSI Transcendence (3.65, 1.36).	The MBSR programme (statistically) significantly improves mindfulness and relaxation measures (particularly positive energy and transcendence) at 8 weeks after baseline. Sustained and even improved long-term effects at 12 months follow-up are shown.

						(4.29, 0.92), SRSI Transcendence (2.64, 0.93).	SRSI mindfulness (4.24, 0.95), SRSI Transcendence (2.40, 0.92).		
Asuero 2014 RCT (2010 - 2012)	Primary health care profes-sionals (41 physician s) in Spain, public system (n=68 total, 47 yrs, 8% male)	8-week MB psycho-educational program: Didactic material, mindful-ness meditation, narrative/a ppreciative enquiry, discussion. Group course 8x2.5h/wk, 1x8h retreat plus homework. (n=43 total, 23 physicians, 48.8yrs)	WLC (n=25 total, 18 physi-cians, 46.9yr)	Mindfuln ess (FFMQ), Empathy (JSPE), also self-report on energy, wellbeing , and activity	Burnout (MBI), mood disturban ce (POMS), intervent ion evaluatio n	Mean and SD. INTERVENTION (n=43). FFMQ Total (129.6, 22.2), FFMQ Observing (26.1, 8.6), FFMQ Describing (28.2, 5.7), FFMQ act with awareness (25.3, 5.9), FFMQ Nonjudging (27.1, 8.0), FFMQ Nonreactivity (21.9, 4.3). JSPE Total (119.5, 13.1), JSPE Perspective taking (54.8, 6.5), JSPE Compassionate care (47.2, 5.1), JSPE standing in pts shoes (13.1, 1.8). CONTROL (n=25). FFMQ Total (120.5, 14.4), FFMQ Observing (24.5, 4.3), FFMQ Describing (26.4, 5.2), FFMQ Act with awareness (23.6, 6.8), FFMQ Nonjudging (26.3, 6.5), FFMQ Nonreactivity (19.8, 2.9). JSPE Total (120.8, 10.1), JSPE Perspective taking (59.8, 7.0), JSPE Compassionate care (47.4, 4.9), JSPE standing in pts shoes (13.6, 0.9).	Time at 8 weeks. Mean and SD. INTERVENTION (n=43) FFMQ Total (141.6, 16.5), FFMQ Observing (30.4, 5.1), FFMQ Describing (28.9, 5.3), FFMQ act with awareness (27.4, 4.7), FFMQ Nonjudging (30.6, 6.2), FFMQ Nonreactivity (24.1, 3.0). JSPE Total (123.0, 9.2), JSPE Perspective taking (56.1, 5.3), JSPE Compassionate care (49.0, 3.9), JSPE standing in pts shoes (13.5, 1.1). CONTROL (n=25) FFMQ Total (121.5, 16.0), FFMQ Observing (24.1, 4.5), FFMQ Describing (26.5, 5.5), FFMQ Act with awareness (23.0, 5.9), FFMQ Nonjudging (27.4, 6.8), FFMQ Nonreactivity (20.2, 3.5). JSPE Total (119.0, 10.7), JSPE Perspective taking (59.6, 6.3), JSPE Compassionate care (46.9, 4.6), JSPE standing in pts shoes (12.5, 3.0).	NA	For mindfulness total there is significant improvement for the intervention (moderate change). For empathy total there is a non-significant increase from pre to post measure in the intervention group, only an increase in compassionate care was statistically significant. Overall mindfulness and empathy (compassionate care) were improved by the programme.

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Barcons 2019 CBA, mixed methods, 2016 - 2017	Primary health care profess-ionals in Spain, public system (n=38).	MTP and IBST group programme plus routine programme . 9 x 1hr per week, (6h Training on psychology, 3h on psychiatry and 1h social work). (n=20)	Rout-ine mental -health sup-port pro-gram for prim-ary care (n=18)	Job Satisfacti on (FR-JS)	Burnout (MBI), Brief Psychiatri c Rating Scale (BPRS), Psychoph armacolo gy use, opinions about mental illness, administr ative and health care indicator s	Median and IQR. INTERVENTION (n=20) FR satisfaction at work (15, 13, 16.25), FR work tension (12, 11, 14), FR Professional competence (5, 4, 6), FR work pressure (12, 10, 12.25), FR professional promotion (12, 9, 13), FR relationship superiors (4, 4, 6.5), FR relationship peers (6, 5.75, 7), FR extrinsic status (5.5, 4 ,6), FR monotony (4.5, 4, 6), FR total (75, 72, 77.5). BPRS total (23.50, 22, 24.25). CONTROL (n=18), FR satisfaction at work (14, 10.5, 16), FR work tension (13.5, 12.25, 16.75), FR Professional competence (4.5, 3, 6), FR work pressure (12, 12, 13), FR professional promotion (11, 10, 13.75), FR relationship superiors (6, 4, 6.75), FR relationship peers (5, 5, 6), FR extrinsic status (5, 5, 6), FR monotony (5, 4, 5.75), FR total (76, 73, 80.5).	Time at 10 months. Median and IQR. INTERVENTION (n=20) FR satisfaction at work (16, 15, 17), FR work tension (13, 11, 16), FR Professional competence (5, 4.5, 6), FR work pressure (12, 10, 12.50), FR professional promotion (12, 10, 13), FR relationship superiors (4, 4, 6.5), FR relationship peers (5, 5, 6), FR extrinsic status (5, 5, 6.5), FR monotony (5, 4, 5), FR total (76, 73, 83). BPRS total (20.50, 19, 22). CONTROL (n=18), FR satisfaction at work (12, 9, 16), FR work tension (15, 1, 16), FR Professional competence (5, 4, 6), FR work pressure (11, 11, 12), FR professional promotion (12, 11, 15), FR relationship superiors (6, 4, 8), FR relationship peers 6, 5, 7), FR extrinsic status (6, 4, 6), FR monotony (4, 3, 5), FR total (77, 75, 78).	NA	No statistically significant changes in job satisfaction (FR). For wellbeing, a statistically significant decrease was seen from pre to post measure in the intervention group, as the BPRS measures negative constructs, this is interpreted as an improvement of wellbeing.
Contratto 2017 NCBA, mixed methods	General medicine physician s in the USA, working part-time (n=7)	1 clerical staff was hired in a GP practice to enter orders (n=7).	NA	Personal-professio nal balance, physician satisfacti on defined as QoL.	Burnout (MBI), physician product-ivity, EMR documen tation. Qual inter-views.	QoL: 14% bad, 14% neutral, 71% good. Balance: 14% Dissatisfied, 43% neutral, 43% good.	Time at 4 months. QoL: 0% bad, 0% neutral, 100% good. Balance: 0% Dissatisfied, 29% neutral, 71% good.		QoI, personal balance, burnout improved. Productivity increased (work relative value unit) per session increased. Reports of feeling more supported, focused on pt care, less stress, less fatigue

Dunn 2007 NCBA, 2000-2005	Primary care in urban USA (n=32, 25 internists 6 family medicine 1 NP)	Organisat-ional inter-vention programme to improve physician wellbeing (quality improve-ment project, participant data-guided interv-ention) (n= 22 to 32 range over the inter-vention period).	NA	Physician satisfacti on (ACP/ASI M)	Burnout (MBI), Quality Work Compete nce (QWC)	2001: 55% of physicians were somewhat or very satisfied with their practice	Time at 24 months. 2003: 84% were somewhat or very satisfied with their practice	Time at 24 months. 2005: 74% were somewhat or very satisfied with their practice.	Intervention ongoing from 2000 onward. Emotional and work-related exhaustion decreased significantly over the study period. QWC measures of organizational health significantly improved initially and remained acceptable and stable during the rest of the study. Satisfaction fluctuated throughout.
Dyrbye 2019 RCT, 2017 - 2018	Physician s in USA (67 generalis ts, 21 subspec-ialists). (n=88, 45.5% male).	6 coaching sessions (3.5h) facilitated by a profess-sional coach over 5mts (n=44).	WLC (n=44)	Resilien-ce (CD-RISC), Global Job Satisfact-ion-12 (subscale of PJSS), work engage-ment (UWES), empower ment at work scale.	Burnout (MBI).	Means and SD. INTERVENTION (n=44), Resilience (31.0, 6.3), Job satisfaction (43.4, 10.7), UWES vigor (3.9, 1.0), UWES Dedication (4.5, 1.1), UWES Absorption (4.2, 1.0), Empowerment at work (55.5, 11.9). CONTROL (n=44), Resilience (30.6, 5.7), Job satisfaction (42.8, 10.6), UWES vigor (4.0, 1.2), UWES Dedication (4.6, 1.0), UWES Absorption (4.1, 1.1), Empowerment at work (57.3, 14.0).	Time at 5 months. Means. INTERVENTION (n=44), Resilience (32.3), Job satisfaction (44.4), UWES vigor (4.1), UWES Dedication (4.6), UWES Absorption (4.1), Empowerment at work (58.2). CONTROL (n=44), Resilience (31.2), Job satisfaction (43.2), UWES vigor (4.2), UWES Dedication (4.7), UWES Absorption (4.2), Empowerment at work (60.3).		Statistically significant improvement for resilience from pre to post intervention, no change in job satisfaction. Burnout, emotional exhaustion decreased. QoL improved. No statistically significant differences in depersonalization, engagement, or meaning in work.
Fortney 2013 nCBA	Family medicine practit-ioners in USA (n=30, 87% family medicine physic-ians)	Shortened MBSR course, 18h total, 14h weekend, 2x 2h plus homework (n=30).	NA	Compas-sion (SCBC), Resili-ence (RS-14).	Burnout (MBI), Depressi on & Anxiety (DASS), Perceive d Stress scale (PSS),	Mean score and Confidence Interval. n=30. Resilience RS-14 (79.9, CI 75.2-84.6), Compassion SCBC (27.6, CI 25.9-29.3).	Time at 4 weeks. Mean score and Confidence Interval. n=28. Resilience RS-14 (82, CI 77.1-86.8), Compassion SCBC (27.4, CI 25.6-29.1).	Time at 9 months. Mean score and Confidence Interval at 9mts. n=23. Resilience RS-14 (81.4, CI 76.2-86.6), Compassion SCBC (28.3, CI 26.5-30.1).	No significant improvement in compassion and resilience over time. Participants had improvements compared with baseline at all follow-up time points for Burnout, depression, anxiety, and perceived stress.

Gardiner 2004 CBA	GPs in metro-politan Australia (n=110)	Cognitive behavioural stress manage-ment course, 1x 3h, 15h over 5 wks. (n=85).	Control group had other devel-opmen tal course s. (n=25)	Work related morale measure (WRM-7), Quality of work life (QoWL-6)	Work related distress measure (WRD-7), General psycholo gical distress (GHQ-12), coping styles.	Means and SD. Total measure without sub-measures given. INTERVENTION (n=86), Work related morale total 31.83 (6.75). QOWL total 25.32 (6.64). CONTROL (n=24), Work related morale total 31.42 (6.19). QoWL total 23.16 (5.86).	Time at 4 weeks. Means and SD. INTERVENTION (n=77), Work-related morale total 34.62 (6.11). QoWL total 28.24 (6.35). CONTROL (n=19), Work related morale total 32.21 (6.73). QoWL total 24.32 (6.36).	Time at 12 weeks. Means and SD. INTERVENTION (n=62), Work related morale total 35.70 (6.01).	Overall the intervention group showed higher scores post intervention than the control group (positive trend). However, no statistically significant change seen for work related morale or QoWL for the intervention group. Only physiological distress significantly lower for intervention group. When looking at those GPs that scored low for morale, there was a 56% reduction pre to post intervention, compared with a 29% reduction in the control group.
Krasner 2009 NCBA	Primary care physici-ans in the USA (n=70, 54% male)	CME programme : Didactic material, mindful-ness meditation, narrative/a ppreciative enquiry, discussion. 28h total. 8 x 2.5hr per week, 1x 7h, 10-month mainten-ance phase (2.5 h/mo). (n=70)	NA	Mindfuln ess (Baer), Empathy (JSPE)	Burnout (MBI), Mood (POMS), Big 5 personali ty factors, Physician belief (PBS)	Mean and CI. Baer mindfulness total (45.2, 95% CI 43.3-47.1, n=60), Baer mindfulness observe (25.6, 95% CI 24.4-26.8, n=60), Baer mindfulness nonreact (19.7, CI 95% 18.7-20.7, n=60). JSPE total (116.2, 95% CI 114.2-118.9, n=60), JSPE compassionate care (48.6, 95% CI 47.5-49.7, n=60), JSPE perspective taking (57.1, 95% CI 55.6 to 58.6, n=60), JSPE standing in pts shoes (10.8, 95% CI 10.4 to 11.5, n=60).	Time at 8 weeks. Mean and CI. Baer mindfulness total (52.9, 95% CI 51 to 54.8, n=59), Baer mindfulness observe (30.6, 95% CI 29.4 to 31.8, n=59), Baer mindfulness nonreact (22.9, CI 95% 21.8 to 23.9, n=59). JSPE total (120.6, 95% CI 118.2 to 123.0, n=59), JSPE compassionate care (49.8, 95% CI 48.7 to 50.9, n=59), JSPE perspective taking (59.1, 95% CI 57.6 to 60.6, n=59), JSPE standing in pts shoes (11.7, 95% CI 11.1 to 12.2, n=59).	Time at 12 months. Mean and CI. Baer mindfulness total (55, 95% CI 53.0 to 56.9, n=56), Baer mindfulness observe (31.1, 95% CI 29.8 to 32.3, n=56), Baer mindfulness nonreact (23.9, CI 95% 22.9 to 24.9, n=56). JSPE total (121.4, 95% CI 119.0 to 123.8, n=56), JSPE compassionate care (50.4, 95% CI 49.3 to 51.5, n=56), JSPE perspective taking (59.7, 95% CI 58.2 to 61.2, n=56), JSPE standing in pts shoes (11.4, 95% CI 10.9 to 11.9, n=56).	Baseline scores and follow-up scores at 15 months are not reported for the purpose of this SR. Over time, all measures for mindfulness, burnout, physician belief, mood and personality improved, the largest effect size was observed for mindfulness at 15 mts.
Linzer 2015 RCT	Primary care clinicians in the USA (n=166, > 80% physic-ians, 47.3 yrs, 48% male)	Interven-tions to improve communica tion; workflow, quality improve-ment (QI). n=83	WLC n=83	Work Control, Satisfacti on (from PWS and MEMO).	Stress, burnout, chaos, intent to leave, varia	Number given in %. INTERVENTION (n=83) High work control (score >2) 96%, High satisfaction (>3) 38.5%. CONTROL (n=83) High work control 13.2%. High satisfaction (>3) 51.8%.	Time at 12 to 18months. Number given in %. INTERVENTION (n=67) High work control (score >2) 4.6%, High satisfaction (>3) 40.0%. CONTROL (n=72) High work control 11.4%, High satisfaction (>3) 45.7%.		Satisfaction improved with workflow interventions, targeted QI projects, communication.

Margalit 2005 RCT	GPs in Israel (n=44)	CPD teaching programs (n=22 for interactive program)	CPD teach-ing progr-am (n=22 didac-tic meth-od)	Attitude to patient care, self-esteem	Knowledge, intention, attitude, burnout	Mean, SD. Self-esteem (72.1, 14.5), Attitude to patient care (34.5, 12.2) (n=44)	Time at 6 months. Mean, SD. Self-esteem (76.5, 12.9), Attitude to patient care (36.5, 19.9) (n=44)	NA	Significant improvement on self-esteem post intervention. The interactive teaching approach improved self-esteem more than the didactic teaching did. No improvement on attitude to patient care.
Montero-Marin 2018 NCBA	GPs in Spain (n=290, 49 yrs, 22.5% males)	Brief blended web-based mindful-ness interven-tion,1x 4h face to face, 8 online sessions (2 weekly sessions over 4 wks) (n=58).	NA	Positive affect (PANAS), awareness (MAAS), resilience (CD-RISC)	Negative affect (PANAS), Burnout subtypes(BCSQ)	Mean, SD. 1 session/wk (n=28): PANAS-pos (32.19, 6.72). MAAS (64, 12.07). CD-RISC (38.96, 8.96). 2 sessions/wk (n=30). PANAS-pos (32.03, 6.38). MAAS (61.77, 13.41). CD-RISC (38.80, 8.58).	Time at 4 weeks. Mean, SD. 1 session/wk (n=28): PANAS-pos (33.44, 5.42). MAAS (66.67, 10.88). CD-RISC (40.19, 5.17). 2 sessions/wk (n=30). PANAS-pos (35.00, 4.91). MAAS (66.37, 11.03). CD-RISC (41.28, 4.32).		Benefits in pos PANAS and MAAS for two or more weekly meditation session. No benefits for one weekly practice. While face to face attendance was good, very high attrition rate for online component.
Pozdnya-kova 2018 NCBA, 2007	Academic general internal medicine clinic (n=6 faculty, n=325 patients).	Clinic sessions with and without a scribe.	NA	Physician workplace satisfaction	Burnout, Time spent on EHR documentation, patient satisfaction with doctor–patient relationship, attitudes towards scribes.	n=6. Number of responses 'agree' or 'strongly agree'. Satisfied with clinic workflow 2/6 (33%). Calm atmosphere in work area 0/6 (0%). Satisfied with quality of patient interactions 5/6 (83%). Satisfied with quality of communication with patient 4/6 (67%).	Time at 3 months. n=6. Number of responses 'agree' or 'strongly agree'. Satisfied with clinic workflow 6/6 (100%). Calm atmosphere in work area 2/6 (33%). Satisfied with quality of patient interactions 6/6 (100%). Satisfied with quality of communication with patient 5/6 (83%).		Of six physicians, 100% were satisfied with clinic workflow post-pilot (vs. 33% pre-pilot), and 83% were satisfied with EHR use post-pilot (vs. 17% pre- pilot). Physician burnout was low at baseline and did not change post-pilot. Mean time spent on post-clinic EHR documentation decreased from 1.65 to 0.76 h per clinic session (p = 0.02).
Rees 2020 NCBA, mixed methods	Rural medical practitioners in Australia (57% GPs). (n=13 total, n=4 qual research, n=7 quant research, 40 yrs, 76.9% male)	MSCR, Mindful Self-Care and Resiliency program (7h face-to-face session) and 3x 1h video-conference (follow up sessions)	NA	Wellbeing (WHO-5), Positive affect (PANAS).	Burnout measure (short version 10-item), psychological strain (GHQ-12)	n=7. Mean. Wellbeing (61.1). Positive affect not reported.	Time at 4 weeks. n=7. Mean. Wellbeing (71.4). Positive affect not reported.		For the WHO-5 wellbeing scale there was no change pre-post for one doctor, deterioration for two doctors with no reliable change, Improvement for four doctors with no reliable change.

Schroeder 2018 RCT, 2014 - 2015	Primary care physici-ans in the USA (n=33, 42.76 yrs, 27% male)	Mindful Medicine Curriculum = modified version of MBSR, 1x 13h & 2x 2h (n=16)	WLC (n=17)	Mindful Attention Awareness (MAAS), Resilience (BRS), Compassion (SCBC),	Stress (PPS), Burnout (MBI), Meditation Practice (MPQ)	Mean, SD. INTERVENTION (n=16) MAAS (3.42, 0.96), BRS (21.62, 4.45), SCBCS (26.31, 4.51). CONTROL (n=17) MAAS (3.32, 0.76), BRS (18.70, 5.13), SCBCS (27.00, 4.97).	Time at 4 weeks. Mean, SD. INTERVENTION (n=15) MAAS (3.62, 0.89), BRS (22.33, 4.74), SCBCS (27.66, 3.22). CONTROL (n=14) MAAS (3.08, 0.76), BRS (19.42, 4.21), SCBCS (26.07, 4.73).	Time at 3 months. Mean, SD. INTERVENTION (n=13) MAAS (4.04, 1.02), BRS (24.15, 5.47), SCBCS (27.84, 4.09). CONTROL (n=13) MAAS (3.18, 0.58), BRS (18.82, 5.32), SCBCS (25.07, 5.85).	Participants in the MMC group reported significant improvements over time for MAAS (also PSS and MBI), whereas in the control group, there were no improvements. There was no significant improvement for resilience or compassion.
Van Vietmar-schen 2018 NCBA, mixed methods, 2015 - 2016	Primary care physic-ians in the Nether-lands (n=54, 87% GPs, 40 yrs, 22% male)	Adapted MBSR programme, weekly group sessions for 8 weeks, 26 hours total.	NA	Self-Compassion (SCS), Self-reflection (GRAS).	Cohen Perceived Stress Scale (PSS)	Mean, SD. SCS (2.9, 0.7, n=50), GRAS (87.6, 7.7, n=44).	Time at 8 weeks. Mean, SD. SCS (3.4, 0.6, n=50), GRAS (90.9, 6.7, n=44).	Time at 6 months. Mean, SD. SCS (3.7, 0.7, n=21), GRAS (90.2, 10.9, n=17).	Significant improvement of self-compassion and self-reflection. Six months after PSS and SCS were still improved. PSS significantly reduced. Qual: awareness, acceptance, peacefulness, and openness improved through intervention.
Beerweij 2016 CBA, mixed methods	General practiti-oner trainers in the Nether-lands (n=50, 54.9 yrs, 66% male)	MBSR training 8x 2.5h, 1x 8h retreat (n=30)	WLC (n=20)	Empathy (JSPE-20), Mindfulness (FFMQ-39)	Work engagement, Burnout (UBOS-C)	Mean, SE. INTERVENTION (n=43) JSPE (117.4, 1.53), FFMQ Total (136.21, 2.23), FFMQ Observing (26.36, 0.69), FFMQ Describing (28.26, 0.9), FFMQ Acting with awareness (27.12, 0.71), FFMQ Non-judging (31.16, 0.81), FFMQ Non-reacting (23.34, 0.57). CONTROL (n=20) JSPE (116.18, 1.92). FFMQ Total (135.48, 2.65), FFMQ Observing (25.85, 0.82), FFMQ Describing (28.33, 0.9), FFMQ Acting with awareness (28, 0.9), FFMQ Non-judging (30.49, 0.98), FFMQ Non-reacting (23.12, 0.7).	Time at 8 weeks. Mean, SE. INTERVENTION (n=43) JSPE (119.35, 1.49), FFMQ Total (143.08, 2.19), FFMQ Observing (28.4, 0.68), FFMQ Describing (29.77, 0.71), FFMQ Acting with awareness (28.1, 0.69), FFMQ Non-judging (32.36, 0.79), FFMQ Non-reacting (24.47, 0.56, 0.57). CONTROL (n=20) JSPE (117.93, 1.98). FFMQ Total (135.45, 2.67), FFMQ Observing (26.33, 0.83), FFMQ Describing (28.2, 0.91), FFMQ Acting with awareness (27.41, .91), FFMQ Non-judging (30.32, 0.99), FFMQ Non-reacting (23.49, 0.7).		Mindfulness skills increased significantly in the MBSR group. Empathy remained the same. The qualitative data indicated that the MBSR course increased their wellbeing and compassion towards themselves and others, including their patients.

Whalley 2008 Panel survey, 2004, 2005	GPs in the UK. 2004: n=2105, mean age 46.9yrs, 66% male. 2005: n=1349, mean age 48.6yrs, 65 % male.	New introduct-ion of pay for perform-ance system happened in 2004 (after the 2004 survey)	NA	JSS WCW	Job pressure, Job design and time pressures	Mean, SD. 2004: JSS total (4.58, 1.39, n=2081)	Time at one year. Mean, SD. 2005: JSS total (5.17, 1.28, n=1345)		Statistically significant improvement in job satisfaction. Job pressure and work hours significantly declined. Most GPs reported that the new contract had increased their income (88%), but decreased their professional autonomy (71%), and increased their administrative (94%) and clinical (86%) workloads.

Key: RCT Randomised Controlled Trial; CBA Controlled Before and After Trial; NCBA Non-Controlled Before and After Trial; QOF Quality and Outcomes Framework; P4P Pay for Performance; NP Nurse Practitioner; WCW Warr Cook Wall; JS Job satisfaction; JSS Job Satisfaction Scale; MBI Maslach Burnout Inventory; JSPE Jefferson Scale of Physician Empathy); POMS Profile Of Mood States; FFMQ Five Facet Mindfulness Questionnaire; SRSI Smith Relaxation States Inventory; NA Not applicable; PBS Physician Belief Scale; SCS Self Compassion Scale (Neff); MPQ Meditation Practice Questionnaire, FR-JS Font Roja Job Satisfaction Questionnaire; BPRS Brief Psychiatric Rating Scale; IQR Inter Quartile Range; ACP College of Physicians; ASIM American Society of Internal Medicine; DASS Depression and Anxiety Scale; PJSS Physician Job Satisfaction Scale (3 dimensions JS, career satisfaction, and specialty satisfaction), 12-item Global Job Satisfaction subscale used; UWES Utrecht Work Engagement Score; CD-RISC Connor-Davidson Resilience Scale; PANAS Positive And Negative Affect Scale, MAAS Mindful attention awareness scale; GHQ General Health Questionnaire; AWS Areas of Work Life Scale (positive measures include control, reward, community, values); BRS Brief Resilience Scale; SCBCS Santa Clara Brief Compassion Scale (an abbreviation of the Sprecher and Fehr's Compassionate Love Scale); MPQ Meditation Practice Questionnaire; UBOS-C Utrecht Burnout Scale for Contractual Occupations (this is the validated Dutch version of the Maslach Burnout Inventory); GRAS Groningen Reflection Ability Scale; EHR Electronic Health Record; MTP Multimodal training programme; IBST integrated brief systemic therapy, PPS Perceived Stress Scale.

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Table 2. Overview of included studies

Intervention level	Intervention type	Study ID (Author, Date)	Study design	Outcome Measure	Within group effect sizes (experiment)	Between group effect sizes (exp vs control group)
Intervention targeted at the individual	Mindfulness	Amutio 2015	RCT	FFMQ	0.78	1.5
		Asuero 2014	RCT	FFMQ, JSPE	0.64, 0.31	0.57, 0.44
		Schroeder 2018	RCT	MAAS, BRS, SCBC	0.63, 0.51, 0.35	0.88, 0.61, 0.73
		Verweij 2016	CBA	FFMQ, JSPE	0.47, 0.20	0.5, 0.02
		Fortney 2013	NCBA	SCBC, RS-14	-0.04, 0.17	-
		Montero-Marin 2018	NCBA	PANAS, MAAS, CD-RISC	0.52, 0.37, 0.37	-
		Rees 2020	NCBA	PANAS, WHO-5	-, 0.52	-
		Van Wietmarschen 2018	NCBA	SCS, GRAS	0.77, 0.46	-
		Krasner 2009	NCBA	FFMQ-2 (BAER), JSPE	1.05, 0.44	-
	Educational training / experiential workshop	Barcons 2019	CBA	FR-JS	-	-
		Gardiner 2004	CBA	WRM-7, QoWL-6	0.43, 0.45	0.3, 0.27
		Margalit 2005	RCT	Self-esteem	-	-
	Coaching	Dyrbye 2019	RCT	CD-RISC, PJSS	0.24, 0.10	0.13, 0.06
Intervention at organisational level	Clerical support / Scribes	Contratto 2017	NCBA	Professional balance, physician satisfaction (QoL)	-	-
		Pozdnyakova 2018	NCBA	Physician workplace satisfaction	-	-
	Organisational improvement programme / QA	Dunn 2007	NCBA	Physician satisfaction	-	-
		Linzer 2015	RCT	Work control, satisfaction	-	-
Intervention at systemic level / policy	Introduction of pay for performance scheme	Allen 2017	NCBA (naturalistic)	Job satisfaction (JSS WCW), Life satisfaction	-	-
		Whalley 2008	NCBA (naturalistic)	JSS WCW	0.44	-

Key: Study design: RCT Randomised Controlled Trial; CBA Controlled Before and After trial; NCBA Non-Controlled Before and After trial.
Outcome measures: JSS WCW Warr Cook Wall Job satisfaction Scale; JSPE Jefferson Scale of Physician Empathy; FFMQ Five Facet Mindfulness Questionnaire; SRSI Smith Relaxation States Inventory; SCS Self Compassion Scale (Neff); FR-JS Font Roja Job Satisfaction Questionnaire; SCBC Santa Clara brief compassion scale (an abbreviation of the Sprecher and Fehr's Compassionate Love Scale); PJSS Physician Job Satisfaction Scale (3 dimensions JS, career satisfaction, and specialty satisfaction), 12-item Global Job Satisfaction subscale used; CD-RISC Connor-Davidson Resilience Scale; PANAS Positive And Negative Affect Scale, MAAS Mindful attention awareness scale; BRS Brief Resilience Scale; GRAS Groningen Reflection Ability Scale; RS-14 Resilience, WHO-5 Wellbeing Index, WRM Work Related Morale Measure. QoWL-6 Quality of Work Life.

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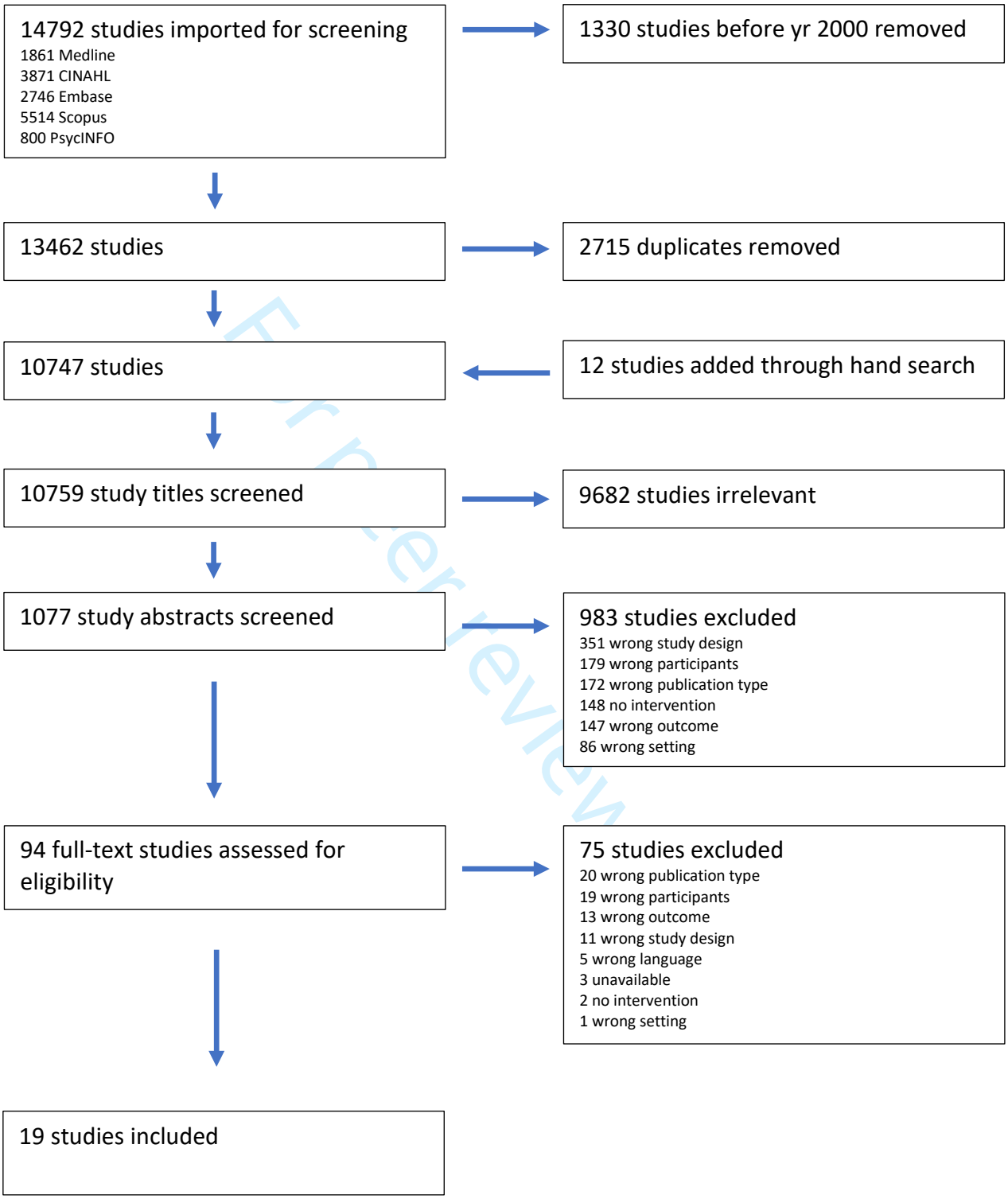
Table 3. Overview of within group and between group Effect Sizes (ES) for several positive outcomes of mindfulness interventions

	Outcome	Mindfulness	Resilience	Compassion	Empathy
Mindfulness interventions (k=9)	Within group ES (k=9)	0.37 – 1.05	0.17 – 0.51	-0.04 - 0.77	0.2 – 0.44
	Between group ES (k=4)	0.5 – 1.5	Only one study 0.61	Only one study 0.73	0.02 - 0.44

There were 9 studies that trialed mindfulness interventions (k=9). These studies utilized a range of different outcomes measures and included tools for assessing mindfulness, resilience, compassion, and empathy. Within group effect sizes (ES) are shown, comparing before and after measures for the intervention.

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Figure 1. PRISMA diagram



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Supplements

- Supplement 1. Medline OVID search strategy
- Supplement 2. Formulae used for Effect Size calculations in Excel for within and between group ES.
- Supplement 3. Effect sizes for outcome measures and sub-measures
- Supplement 4. Outcome measures utilized in the included studies.

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Supplement 1. Medline OVID search strategy

1. general practitioners/

2. "general practitioner*".mp.

3. (GP or GPs).mp.

4. exp General Practice/

5. Family Practice/ or Primary Health Care/

6. "family practitioner*".mp.

7. "primary care practitioner*".mp.

8. physicians, family/ or physicians, primary care/

9. "family medicine".mp.

10. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9

11. ("psychological well being" or "psychological wellbeing").mp.

12. (wellbeing or "well being").mp.

13. ("cognitive well being" or "cognitive wellbeing").mp.

14. flourishing.mp.

15. ((job or work) adj3 (satisf* or engage* or motivat*)).mp.

16. life satisfaction*.mp.

17. Job Satisfaction/

18. resilien*.mp.

19. Resilience, Psychological/

20. burnout.mp.

21. Burnout, Professional/

22. Compassion fatigue/

23. compassion fatigue.mp.

24. joy.mp.

25. "joy in practice".mp.

26. 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25

27. Motivational interviewing/

28. Mindfulness/

29. Mediation/

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30. mindful*.mp.

31. meditat*.mp.

32. Self-Care/

33. ("self care" or "self-care").mp.

34. Motivation/

35. Professional autonomy/

36. ("self determination" or "self-determination").mp.

37. exp Randomised Controlled Trial/ or Clinical Trial/

38. trial.mp.

39. Early Medical Intervention/ or intervention.mp.

40. Counseling/

41. Course?ling.mp.

42. Mentoring/

43. mentor*.mp.

44. Education/

45. education.mp.

46. workshop*.mp.

47. training.mp.

48. coaching.mp.

49. facilitat*.mp.

50. 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49

51. 10 and 26 and 50

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Supplement 2. Formulae used for Effect Size calculations in Excel for within and between group ES.

Within-group:

Mean_change_WG = Mean_Exp_post – Mean_Exp_pre

SD* pooled _WG = SQRT(((N_Exp_post-1) * SD_Exp_post^2) + (N_Exp_pre-1) * SD_Exp_pre^2)) / (N_Exp_post + N_Exp_pre – 2))

ES_WG = Mean_change_WG / SD* pooled_WG

Between-group:

Mean_change_BG = (Mean_Exp_post – Mean_Exp_pre) – (Mean_Ctrl_post – Mean_Ctrl_pre)

SD* pooled_BG = SQRT(((N_Ctrl_pre-1) * SD_Ctrl_pre^2) + (N_Exp_pre-1) * SD_Exp_pre^2)) / (N_Ctrl_pre + N_Exp_pre – 2))

ES_BG = Mean change_BG / SD* pooled_BG

Legend. ES = Effect size, WG = Within group, BG = Between group, Exp = experimental group, Ctrl = control group, N = sample size, pre = pre-intervention measure, post = post-intervention measure, SQRT = square root, ^2 = raised to the power of two.

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Supplement 3. Effect sizes for outcome measures and sub-measures

Name, n pre/post	Measure	Within group absolute mean change (post - pre experiment)	SD* Exp pooled	ES Hedges' g Exp (Delta mean/ SD* pool)	Between group absolute mean change (experiment – control group)	SD* Baseline pooled	ES Hedges' g (Delta Mean / SD* pool)
With control group							
AMUTIO*	FFMQ	0.37	0.48	0.78	0.54	0.36	1.51
n=21/21	Observe	0.65	0.62	1.05	0.84	0.75	1.12
	Describe	0.25	0.67	0.37	0.31	0.63	0.49
	Awareness	0.32	0.77	0.42	0.34	0.79	0.43
	Non-judgement	0.36	0.65	0.55	0.39	0.58	0.67
	Non-reactive	0.29	0.57	0.51	0.26	0.54	0.48
	SRSI-3 Relaxation	0.54	0.57	0.95	0.46	0.52	0.89
	SRSI-3 Pos energy	0.71	0.74	0.97	0.71	0.63	1.13
	SRSI-3 Mindfulness	0.71	0.81	0.88	0.76	0.91	0.84
	SRSI-3 Transcend	0.6	0.93	0.65	0.84	0.88	0.96
VERWEIJ**	FFMQ	6.87	14.49	0.47	6.9	13.82	0.50
n=43/43 exp,							
n=20/20 control	Observe	2.04	4.49	0.45	1.56	4.28	0.36
	Describe	1.51	4.69	0.32	1.64	4.52	0.36
	Awareness	0.98	4.59	0.21	1.57	4.47	0.35
	Non-judgement	1.2	5.25	0.23	1.37	5.04	0.27
	Non-reactive	1.13	3.71	0.30	0.76	3.56	0.21
	JSPE total	1.95	9.90	0.20	0.2	9.61	0.02
ASSUERO	FFMQ	12.40	19.50	0.64	11.3	19.65	0.57
n=43/43 exp							
n=25/25 control	Observe	4.30	7.07	0.61	4.7	7.33	0.64
	Describe	0.70	5.50	0.13	0.6	5.52	0.11
	Awareness	2.10	5.33	0.39	2.7	6.24	0.43
	Non-judgement	3.50	7.05	0.50	2.4	7.35	0.33
	Non-reactive	2.20	3.71	0.59	1.8	3.85	0.47
	JSPE total	3.50	11.32	0.31	5.3	12.10	0.44
	Perspective	1.30	5.88	0.22	1.5	6.62	0.23
	Compassion	1.80	4.54	0.40	2.3	5.03	0.46
	In patients shoes	0.40	1.49	0.27	1.5	1.54	0.98
DYRBYE	CD-RISC-10	1.30	5.44	0.24	0.7	5.46	0.13
n=40/40 exp							
n= 39/39 control	PJSS	1.00	9.94	0.10	0.6	10.11	0.06
	UWES vigor	0.30	0.94	0.32	0.2	1.04	0.19
	UWES dedication	0.10	0.87	0.11	-0.1	0.96	-0.10
	UWES absorption	-0.10	0.88	-0.11	-0.2	1.00	-0.20
	Empowerment at work	2.7	11.29	0.24	-0.3	12.56	-0.02
GARDINER	WRM-7 total	2.79	6.46	0.43	2	6.63	0.30
n=86/77 exp,							
n=24/19 control	QoWL-6 total	2.92	6.50	0.45	1.76	6.48	0.27
SCHROEDER	MAAS	0.62	0.99	0.63	0.76	0.86	0.88
N=16/13 exp,							
n=17/13 control	BRS	2.53	4.93	0.51	2.95	4.81	0.61
	SCBC	1.53	4.33	0.35	3.46	4.75	0.73
MARGALIT***							
n=44/44	Self-esteem	4.4	13.72	0.32			

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Without control group					
MONTERO-MARIN n=30/30	PANAS 1 session/wk	1.25	6.10	0.20	
	PANAS 2 session/wk	2.97	5.69	0.52	
	MAAS 1 session/wk	2.67	11.49	0.23	
	MAAS 2 session/wk	4.59	12.28	0.37	
	CD-RISC total 1 session/wk	1.23	7.31	0.17	
	CD-RISC total 2 session/wk	2.48	6.79	0.37	
VON WIETMARSCHEN n=50/50	SCS	0.5	0.65	0.77	-
	GRAS	3.3	7.22	0.46	-
FORTNEY n=30/28	RS-14	2.1	12.28	0.17	-
	SCBC	-0.2	4.54	-0.04	-
KRASNER n=60/59	JSPE total	4	9.16	0.44	-
	Perspective	2	5.79	0.35	-
	Compassion	1.6	4.24	0.38	-
	In patients' shoes	0.8	2.12	0.38	-
	Mindfulness 2 factor FFMQ (BAER) total	7.7	7.33	1.05	-
	BAER observe	5	4.63	1.08	-
	BAER non-react	3.2	3.95	0.81	-
REES n=7/7	WHO-5 (WB)	10.3	19.79	0.52	-
	PANAS-pos	-			-
WHALLEY n=2081/1345	JSS WCW	0.59	1.35	0.44	-

Notes: Amutio*, no information given on pre-post sample size, assumption made that there was no attrition. Verweij**, wait list control participants were included in both intervention and control groups. *** Margalit only pooled group (experimental and control) analysis provided. For the FFMQ Verweij and Assuero have much larger numbers than Amutio, this is presumably due to reporting of the total score count as opposed to the total mean scores.

Outcome measures: JSS WCW Warr Cook Wall Job satisfaction Scale; JSPE Jefferson Scale of Physician Empathy; FFMQ Five Facet Mindfulness Questionnaire; Baer (2 item version of FFMQ); SRSI Smith Relaxation States Inventory; SCS Self Compassion Scale (Neff); FR-JS Font Roja Job Satisfaction Questionnaire; SCBC Santa Clara brief compassion scale (an abbreviation of the Sprecher and Fehr's Compassionate Love Scale); PJSS Physician Job Satisfaction Scale (3 dimensions JS, career satisfaction, and specialty satisfaction), 12-item Global Job Satisfaction subscale used; CD-RISC Connor-Davidson Resilience Scale; PANAS Positive And Negative Affect Scale, MAAS Mindful attention awareness scale; BRS Brief Resilience Scale; GRAS Groningen Reflection Ability Scale; RS-14 Resilience, WHO-5 Wellbeing Index, WRM Work Related Morale Measure. QoWL-6 Quality of Work Life.

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Supplement 4. Outcome measures utilized in the included studies.

Acronym	Outcome measure	No of studies used in
ACP/ASIM	Physician satisfaction (Am College of Phys / Am Soc of Int Med)	1
BAER	Baer mindfulness scale (2 item version of FFMQ)	1
BRS	Brief resilience scale	1
CD-RSIC-10	Connor-Davidson Resilience Scale	2
FFMQ	Five Facet Mindfulness Questionnaire	3
FR-JS	Font Roja Job Satisfaction scale	1
GRAS	Groningen Reflection Ability Scale	1
JSPE	Jefferson Scale of Physician Empathy	3
JSS WCW	Job Satisfaction Scale Warr-Cook-Wall	2
LS	Life satisfaction	1
MAAS-15	Mindful attention awareness scale	2
PANAS	Positive and negative affect scale	2
PJSS	Physician Job satisfaction scale	1
PWS (MEMO)	Physician Workplace Satisfaction	2
QoWL-6	Quality of work life	1
RS-14	Resilience	1
SCBC	Santa Clara brief compassion scale	2
SCS	Self compassion scale (Neff)	1
SRSI-3	Smith Relaxation states	1
UWES	Utrecht work engagement score	1
WHO-5	Wellbeing Index	1
WRM-7	Work related morale measure	1
	Self-esteem	1

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The PRISMA for Abstracts Checklist

TITLE	CHECKLIST ITEM	REPORTED ON PAGE #
1. Title:	Identify the report as a systematic review, meta-analysis, or both.	1
BACKGROUND		
2. Objectives:	The research question including components such as participants, interventions, comparators, and outcomes.	2
METHODS		
3. Eligibility criteria:	Study and report characteristics used as criteria for inclusion.	2-4
4. Information sources:	Key databases searched and search dates.	
5. Risk of bias:	Methods of assessing risk of bias.	
RESULTS		
6. Included studies:	Number and type of included studies and participants and relevant characteristics of studies.	4-8
7. Synthesis of results:	Results for main outcomes (benefits and harms), preferably indicating the number of studies and participants for each. If meta-analysis was done, include summary measures and confidence intervals.	
8. Description of the effect:	Direction of the effect (i.e. which group is favoured) and size of the effect in terms meaningful to clinicians and patients.	
DISCUSSION		
9. Strengths and Limitations of evidence:	Brief summary of strengths and limitations of evidence (e.g. inconsistency, imprecision, indirectness, or risk of bias, other supporting or conflicting evidence)	8-12
10. Interpretation:	General interpretation of the results and important implications	
OTHER		
11. Funding:	Primary source of funding for the review.	13
12. Registration:	Registration number and registry name.	